

Fintech Key-Phrase: A New Chinese Financial High-Tech Dataset Accelerating Expression-Level Information Retrieval

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Expression-level information extraction is a challenging task in natural language processing (NLP), which aims to retrieve crucial semantic information from linguistic documents. However, there is a lack of up-to-date data resources for accelerating expression-level information extraction, particularly in the Chinese financial high technology field. To address this gap, we introduce Fintech Key-Phrase: a human-annotated key-phrase dataset for the Chinese financial high technology domain. This dataset comprises over 12K paragraphs along with annotated domain-specific key-phrases. We extract the publicly released reports, Chinese management's discussion and analysis (CMD&A), from the renowned Chinese research data services platform (CNRDS) and then filter the reports related to high technology. The high technology key-phrases are annotated following pre-defined philosophy guidelines to ensure annotation quality. In order to better understand the limitations and challenges in the purposed dataset, we conducted comprehensive noise evaluation experiments for the Fintech Key-Phrase, including annotation consistency assessment and absolute annotation quality evaluation. To demonstrate the usefulness of our released Fintech Key-Phrase in retrieving valuable information in the Chinese financial high technology field, we evaluate its significance using several superior information retrieval systems as representative baselines and report corresponding performance statistics. Additionally, we further applied ChatGPT to the text augmentation approach of the Fintech Key-Phrase dataset. Extensive comparative experiments demonstrate that the augmented Fintech Key-Phrase dataset significantly improved the coverage and accuracy of extracting key phrases in the finance and high-tech domains. We believe that this dataset can facilitate scientific research and exploration in the Chinese financial high technology field. We have made the Fintech Key-Phrase dataset and the experimental code of the adopted baselines accessible on Github: https://github.com/albert-jin/Fintech-Key-Phrase. To encourage newcomers to participate in the financial high-tech domain information retrieval research, we have developed a series of tools, including an open website[1] and corresponding real-time information retrieval APIs.[2]

[1] https://albert-jin.github.io/FintechKP-frontend/
[2] https://31863ew564.zicp.fun/information_retrieval

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CCS Concepts: • **Information systems** → **Specialized information retrieval**; *Retrieval effectiveness*; *Data extraction and integration*; **Content analysis and feature selection**; • **Applied computing** → *Electronic data interchange;*

Additional Key Words and Phrases: Information retrieval, expression-level information extraction, financial high technology field, Chinese management's discussion and analysis, ChatGPT-based data augment

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1 INTRODUCTION

In **natural language processing** (**NLP**), **Information retrieval** (**IR**) plays a crucial role and encompasses various research directions, including **named entity recognition** (**NER**) [42] and **aspect-based sentiment analysis** (**ABSA**) [20, 21]. Within information retrieval, expression-level information extraction is a sub-task that focuses on extracting interesting words, phrases, or even paragraphs from large-scale textual documents [24]. This task is similar to NER and requires the machine to achieve a comprehensive semantic understanding, accurately identifying whether each atomic expression, known as the minimum semantic unit, in a sentence contains interesting key information. The results of expression-level information extraction can benefit various downstream NLP tasks, such as **knowledge graph retrieval-based question answering** (**KGQA**) [20] and document topic classification [44]. As a result, expression-level information extraction has emerged as one of the key challenges in linguistic processing problems in recent years.

Despite the recent success in expression-level information extraction, there still remains a dearth of specific-domain information extraction datasets to fully explore its potential in research and application, which hampers the further development of textual semantic analysis. Specifically, to our knowledge, there are a lack of publicly available data sources for the expression-level information extraction task in the Chinese financial high-tech field. To address these gaps and foster in-depth research, we introduce a novel dataset called Chinese **Fin**ancial high-**tech** based **Key-Phrase** (Fintech Key-Phrase), which serves as the latest benchmark for expression-level information extraction in Chinese financial high-tech domain.

(1) Firstly, we extract the original corpus from the renowned **Chinese research data services platform** (**CNRDS**),[3] which consists of nearly one hundred thousand annual financial corporation reports.
(2) Next, we employ an enterprise category mapping strategy to identify and filter the sections related to high technology enterprise from all annual financial reports. These filtered enterprise annual reports serve as the source documents for generating our Fintech Key-Phrase dataset.
(3) To ensure the reliability of the dataset, the annual reports are manually annotated by a professional team of crowd-sourced annotators with financial background following pre-defined annotation rules. This approach avoids relying on automatic labelling based on fixed matching patterns.

In this work, we make the following major contributions:

— We present a novel dataset, Fintech Key-Phrase for the expression-level information retrieval task in the domain of Chinese financial high-tech. This dataset is derived from the publicly

[3] https://www.cnrds.com/

released **Chinese management's discussion and analysis** (**CMD&A**). To the best of our knowledge, and it is the largest reliable Chinese benchmark available for the expression-Level Information Extraction, containing over 1.2K human-annotated instances;
— We performed a comprehensive noise evaluation for the purposed Fintech Key-Phrase dataset, which involved evaluating the level of annotator consistency and the absolute accuracy of annotations. With the inclusion of several typical annotation cases, this work provided valuable insights into the dataset, leading to a more comprehensive understanding of its characteristics;
— To emphasize the significance of Fintech Key-Phrase in expression-Level Information Extraction, we conduct comprehensive experiments using **state-of-the-art** (**SOTA**) approaches. The experimental results validate the dataset's suitability as a solid baseline for future research in Information Extraction;
— Motivated by the prominent text generalization ability of ChatGPT, we firstly introduced ChatGPT in the data augmentation process of Fintech Key-Phrase. Using ChatGPT, we expanded the original dataset by over three times. Through comparative experiments, we verified the effectiveness of this text augmentation strategy in improving the information retrieval performance. To the best of our knowledge, this also is the first exploration and investigation of applying ChatGPT to the expression-level information retrieval tasks in the Chinese financial and high-tech domains;
— For reproducibility, we have made the complete original corpus (in Excel format), the annotated Key-Phrase dataset (in JSON format), and the experimental evaluation scripts (in Python) publicly available at https://github.com/albert-jin/Fintech-Key-Phrase/;
— In order to encourage newcomers to participate in the Chinese financial high-tech information retrieval field, we have developed a series of tools, including an open website[4] and corresponding real-time information retrieval APIs.[5]

The remainder of the article is organized as follows. Section 2 introduces the related research and widely used benchmarks for the expression-level information extraction task. In Section 3, we provide the main motivation behind our work and elaborate a detailed description of the Fintech Key-Phrase dataset construction. Additionally, we conduct comprehensive noise evaluation experiments to assess the quality of the dataset. Section 4 discusses the experimental results of several SOTA methods using our benchmark, which can be considered as new baselines for the Fintech Key-Phrase dataset. In Section 5, we extensively discuss the experimental process of text augmentation for our Fintech Key-Phrase using ChatGPT. In Section 6, we provide a comprehensive introduction on how to utilize the tools we have released. Furthermore, we demonstrate the significance of our proposed benchmark through detailed case studies and error analyses in Section 7. Finally, in Section 8, we summarize the contributions of our research and offer a brief perspective on future directions.

2 RELATED WORK

2.1 Expression-Level Information Extraction

Expression-level information extraction, a widely used text mining technique, has recently garnered significant attention due to the growing demand for valuable knowledge extracted from vast digital document collections [25, 40, 41]. As the name implies, this technique involves automatically identifying valuable domain-specific information, such as phrases, proper nouns, and

[4]https://albert-jin.github.io/FintechKP-frontend/
[5]https://31863ew564.zicp.fun/information_retrieval

event descriptions, from large-scale unstructured documents [38]. These extracted key phrases often convey richer valuable semantic information compared to function words. Consequently, the extracted results can effectively facilitate various downstream NLP tasks, including **knowledge graph-based question answering** (**KGQA**) [20] and ABSA [21, 43]. In recent years, numerous studies have made remarkable progress, and achieved superior performance in the field of expression-level information extraction [38].

Early approaches to expression-level information extraction relied on manual creation of matching patterns and pre-/post-processing rules. Eddy et al. [7] introduced improvements by incorporating **Hidden Markov Models** (***HMMs***) and utilizing external information such as deterministic, morphological, and semantic triggers to enhance entity extraction performance. In the domain of NER in biological medical field, Zhou et al. [46] presented *PowerBioNE*, a system based on HMMs and **k-nearest neighbor** (**k-NN**) algorithm, which improved post-processing patterns for automatic extraction of interesting entities from sentences. However, these rule-based methods heavily rely on feature engineering, necessitating the design of numerous expert-crafted features, which is time-consuming and lacks effective domain transferability.

In recent years, with the rapid advancement of **artificial intelligence** (**AI**), **deep neural network** (**DNN**)-based approaches have demonstrated significant superiority over traditional **machine learning** (**ML**)-based approaches. Zhiheng et al. [17] introduced the '*BiLSTM-CRF*' network, which transforms the initial information extraction task into a simple sequence tagging problem. Fei et al. [8] proposed a novel multitask learning method based on a dispatched attention module. They first encoded an input sentence using a Bi-LSTM layer to obtain contextualized representations. Then, an attention module incorporating position and syntax knowledge was utilized to capture relevant features. Finally, the attention representation of each task is allocated to predict the target boundaries. Subsequently, Nguyen et al. [30] proposed a joint extraction framework for information extraction called '*Multi-Stage Attentional U-Net*'. This framework effectively extracts valuable information from large-scale unstructured texts, such as receipts and cheques. The model utilizes a specialized multi-stage encoder-decoder structure, leveraging self-attention and box convolution operations, and achieves remarkable performance surpassing many competitive baselines [16]. As we know, creating token-level labels can be time-consuming in most information extraction methods, particularly those relying on such labels to extract target phrases from raw sentences. To address this challenge, Palm et al. [31] proposed an E2E pointer network with attention. This network eliminates the need for expensive token-level labels by breaking long-term dependencies and achieves competitive performance compared to neural network-based baselines.

2.2 Domain-Specific Benchmarks

Despite the significant progress made at the model level in the expression-level information extraction task, there are also numerous widely used benchmark datasets.

Firstly, we will describe several widely used benchmark datasets in open-domain information extraction. In nested Nested NER, Ringland et al. [33] presents the **Nested Named Entity** (***NNE***) corpus of named entities and numerical and temporal expressions for extracting people, locations, and organizations from natural language texts in English. It is derived from the *Wall Street Journal*'s ***Penn Treebank*** (**PTB**) block and annotated based on twelve high-level principles. The annotations resolve difficult decisions and cover 118 hierarchical fine-grained entity types and nesting rules, including capitalized mentions of entities, numerical expressions, and temporal expressions. The dataset contains approximately 279,000 annotated entity mentions and more than 100 entity categories. It enables intelligent systems to automatically exploit interesting entity targets, benefiting various downstream NLP tasks such as *expression-level information extraction* and *document-level reading comprehension*. In the German NER field, GermEval 2014 [2] is a well-known German

NER benchmark built on the German News and Wikipedia Corpora. The dataset consists of over 3,000 manually labeled sentences, corresponding to more than 5,900 words from Wikipedia and online news in German. It contains over 41,000 named entities, including embedded named entities, derivations, and concatenated named entities with other words. The dataset covers 12 named entity classes, including *Person*, *Location*, *Organisation*, and *Others*, and is fully available on the official website.[6]

Furthermore, many domain-specific benchmark datasets have been constructed by researchers to advance the development of domain-specific information retrieval tasks. In the biological chemical domain expression-level information extraction, Li et al. [24] proposed a large-scale chemical patent analysis benchmark for the ChEMU 2022 campaign. This benchmark focuses on promoting drug design technique and material manufacturing analysis. The task includes three sub-tasks: Anaphora Relation Identification, which involves resolving anaphoric dependencies from textual corpora; Named Entity Recognition, which aims to identify valuable information, such as reaction temperatures, time consumption, drug compounds, and their chemical types; and Event Extraction which recognizes a series of related individual events occurring from a chemical reaction to an end product [10, 37]. In the mental health field, Soumitra et al. [13] have re-annotated the *CEASE-v2.0* dataset [12] and introduced a new benchmark dataset focused on emotion-cause annotated suicide. This dataset enhances the performance of recognizing emotion-cause information conveyed in suicide notes. Additionally, they contribute a substantial volume of multilingual suicide notes corpus and present a standardized strategy for emotion cause annotation.

In addition to the aforementioned datasets, there are numerous open-source information retrieval systems available. In the **foreign exchange** (**Forex**) markets, Lee et al. [23] have proposed an interactive web-based system called INForex. This system displays Forex plots alongside related financial news, unveiling valuable relationships and insights between textual news data and Forex data. It serves as a valuable tool for analysis and decision-making for professionals in the field, while also benefiting ordinary investors.

3 FINTECH KEY-PHRASE: A NEW DATASET FOR EXPRESSION-LEVEL INFORMATION RETRIEVAL

3.1 Main Motivation

With the rapid advancement of science, technology, and industrialization, the 21st century has witnessed the ongoing trend of technological revolutions and economic globalization. In recent decades, the financial high technology field has garnered significant attention from both industries and academic communities. It is well-known that Chinese high technology techniques, which have progressed alongside the rapid development of the financial market, provide a wealth of valuable experiences that can promote global technological advancement. An intelligent financial analysis system can effectively assist investors in various investment activities, such as *stock price forecasting*, *corporate risk assessment*, and *monitoring stock price volatility*). Efficiently extracting relevant information in the financial high technology domain remains a crucial area of research.

Currently, at the NLP methodological level, most improvements are based on enhancing the Transformer architecture. The dataset development and method advancements should be intricately intertwined, jointly driving the progress in NLP. However, innovation in model structure and algorithms seems to have reached a significant "bottleneck". For mainstream neural network-based (NN-based) supervised learning approaches, the availability of sufficient annotated corpora directly impacts the final performance of the NN-based models. Unfortunately, due to the nascent

[6] https://sites.google.com/site/germeval2014ner/data

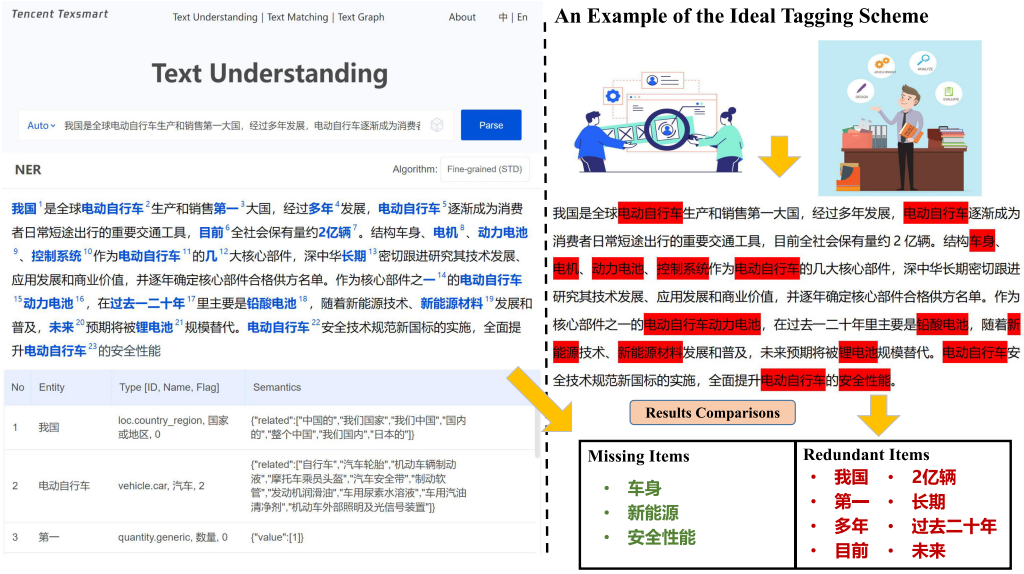


Fig. 1. Comparisons of information extraction results between the open-domain text understanding interface '*TexSmart*' and our ideal tagging results, which are annotated by financial experts.

stage of domain-specific information extraction, there is currently a lack of available data sources in the financial high technology domain. This severe data deficiency has created a substantial gap in specific domain-related research. As a result, there is a growing emphasis on the development of newer, more challenging datasets with richer application scenarios and the definition of new task settings.

The AI & NLP team at '*TENCENT*', a leading internet technology corporation, has developed '*TexSmart*' [26], a powerful NLP and analysis tool. TexSmart encompasses various NLP tasks, including semantic extension, Nested NER, and Fine Grained Semantic Expression for specific entities types. For more information, please visit the official website at https://texsmart.qq.com/.

While '*TexSmart*' provides online APIs[7] for open-domain expression-level information retrieval task, its performance may be limited in domain-specific scenarios such as the financial high-tech domain. To investigate this potential limitation, we conducted a thorough comparison between the information retrieval results obtained from '*TexSmart*' and the expert-annotated standard in the financial domain. As depicted in Figure 1, we observed that '*TexSmart*' produced a significant number of *Missing Items* (i.e., keywords that should have been included) and *Redundant Items* (i.e., keywords that should not have been included) compared to the experts' ideal annotations. For instance, in this comparative keyword analysis, '*TexSmart*' failed to identify the keyword phrases '新能源' (New Energy) and '安全性能' (Safety Performance) that are relevant to the new energy vehicle industry. Additionally, it excessively extracted some phrases such as '长期' (long-term), '目前' (currently), and '未来' (future), which hold limited significance for subsequent industry analysis and decision-making. Such inaccuracies can adversely affect downstream NLP tasks like document topic recognition [44] and paragraph-level reading comprehension [10, 40]. Consequently, it is crucial to develop an information retrieval dataset specifically tailored for the financial high-tech domain, aiming to enhance the performance of Information Extraction models.

[7]TexSmart: https://texsmart.qq.com/en

To address these gaps, we have created a substantial human-annotated dataset, called Fintech Key-Phrase, for the expression-level information extraction task. Our aim is for this benchmark to play a crucial role in uncovering valuable underlying knowledge regarding the development patterns in the Chinese financial high technology field.

3.2 Dataset Construction Guidelines

The construction of our dataset involved a series of preparation procedures. Initially, we collected a large volume of unstructured corpora from the Chinese Internet; Subsequently, we implemented a novel filtering strategy to extract domain-specific texts related to the high technology industry. Following a set of predefined labelling guidelines, we engaged crowd-sourced workers who are all Chinese financial-domain experts to annotate the interesting phrases within the documents. These annotation experts are all from the School of Business and Finance at Nanjing University,[8] possessing a strong foundation in financial knowledge and a specialized understanding of the field.

3.2.1 Corpus Source Collecting. Prior to the manual annotations, the strategy employed for corpora collection was a crucial step that directly influenced the annotating procedures and the final quality of our Fintech Key-Phrase dataset. The corpus collection process can be divided into two primary steps: *Extraction from CMD&A*, and *Fintech Domain Filtering*.

Extraction from CMD&A. The Internet houses a vast amount of Chinese financial market data, including company trade records, enterprise business condition announcements, and corporate annual reports. Notably, the CNRDS, a well-known platform, has archived an extensive collection of historical information on most Chinese public companies. Specifically, the CNRDS platform encompasses hundreds of millions of multidimensional textual pieces of information for comprehensive analysis, such as research reports, stock volatility records, credit or annual reports, and legal documents. Our focus is on Chinese annual enterprise reports, also known as CMD&A.[9] CMD&As provide a combination of business information (e.g., Financial Statements) and non-business information (e.g., Corporate Administrations and Core Techniques), along with the professional perspectives of these corporations. We request the official API and obtained the CMD&A documents of Chinese financial enterprises from the 2008 to the 2022 (a span of 14-years until the present), resulting in a corpus containing millions of annual enterprise reports.

Fintech Domain Filtering. After obtaining a substantial collection of CMD&A documents, our next step is to accurately filter out the high technology companies from the original documents. We discovered that the **China Securities Regulatory Commission** (**CNSRC**)[10] has established the **China National Economic Industry Classification Guidelines** (**CNEICG**).[11] The Guidelines provide pre-defined industrial classifications for all Chinese public companies, including the various industrial categories and their corresponding indexes. For instance, examples of such classifications include '核燃料加工' *Nuclear-Fuel Processing* [No. 253] and '航空器制造' *Spacecraft Manufacturing* [No. 3726]. We leverage these classification guidelines to effectively filter out the CMD&A documents that are related to the high technology sector from the original corpus.

Here, we briefly introduce the statistics of the filtered CMD&A documents:

— The high-tech CMD&A annual reports consist of more than 16,600 documents, recording the annual business reports of up to 2,692 different companies;

[8] https://nubs.nju.edu.cn/main.htm
[9] https://www.cnrds.com/Home/index#/FeaturedDatabase/DB/CMDA
[10] http://www.csrc.gov.cn/
[11] http://www.csrc.gov.cn/csrc/c100103/c1452025/content.shtml

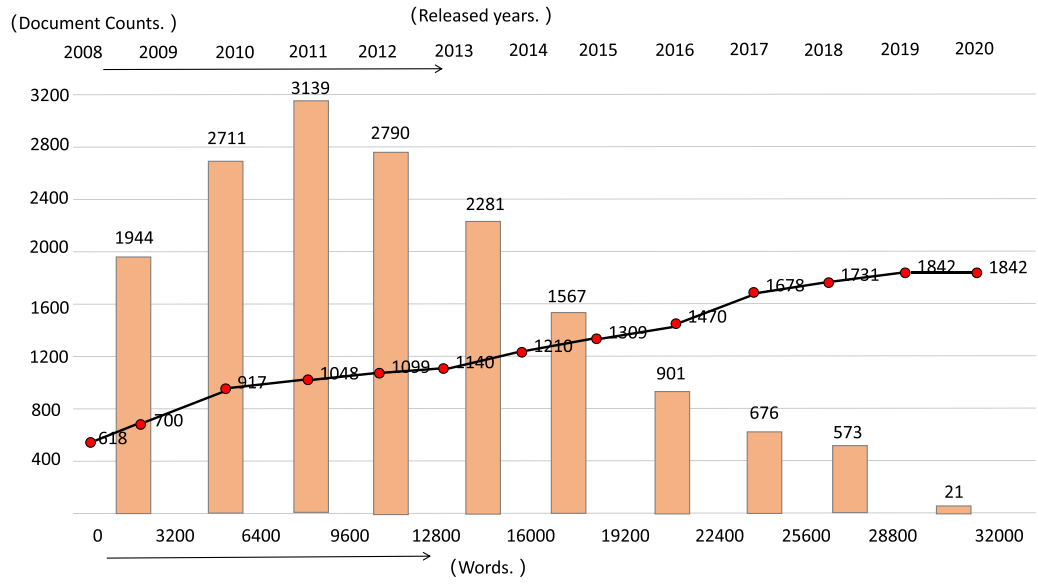


Fig. 2. Statistics of the extracted financial high-tech enterprises CMD&A reports.

— On average, the high technology CMD&A documents contain approximately 11,171 words. The length of the documents ranges from a minimum of 115 words to a maximum of 32,006 words;
— The statistics regarding the document lengths and release times in different intervals are illustrated in Figure 2.

3.2.2 Key-Phrase Annotation. Different from automatic construction, human annotation can endow datasets with higher recall and precision while reducing unexpected noise. We have formulated several primary labelling schemes to circumscribe the scope of the interesting phrases that need to be annotated. In other words, we recruited crowd-sourced annotators to mark the target texts following these regulations in a set of specific-domain documents.

— **Word Meaning Rules.** Annotators should pay greater attention to phrases related to enterprise products, techniques, attributes, and events that leads to better insights into the nature of the enterprise, its recent condition, and the overall theme of the document;
— **Suffixes Exclusion Rules.** Annotators should exclude common-used suffixes (e.g., XXX'制度', *XXX institution*; XXX'技术', *and XXX technique*) from each key-phrase due to semantic redundancy. For example, in the sentence '锂离子电池产品最近几年成为售卖主力(Lithium battery products have become the main sales force this year)', we expect to extract '锂离子电池(Lithium battery)' rather than '锂离子电池产品(Lithium battery products)';
— **Conjunction Exclusion Rules.** Regarding conjunctions rules, annotators should omit conjunctions between two key-phrases as they lack semantic meaning. For example, in the sentence '景峰制药启动了研发中心项目、生物药项目、脂质体项目的设计与建设(Jingfeng Pharmaceutical started the design and construction of biological drug projects, liposome projects, and R&D centre projects)', we expect to extract the three key-phrases separately: '生物药项目(drug projects)', '脂质体项目(liposome projects)', and '研发中心

Table 1. Examples of Key-phrase Annotations that Follow the Labelling Scheme Described Above

	Sentence	Key-Phrases
S1	完成"面向下一代互联网感知-智能流量控制设备产业化项目"	互联网感知　智能流量控制
S2	天融信开展永磁环保柜、永磁环网柜及环网柜等开发项目	永磁环保柜　永磁环网柜　环网柜
S3	哈工易科加强物联网市场基础建设,加大研发人工智能机器人	物联网　人工智能机器人

项目(R&D centre projects)', rather than extracting the continuous text '研发中心项目、生物药项目、脂质体项目(drug projects, liposome projects, and R&D centre projects)';

— **Universal Semantic Rules.** Annotators should skip labelling micro-aspect words, such as proprietary phrases created by the enterprise itself. Instead, annotators are encouraged to extract explicit phrases that contain universal semantic information and appear more frequently. For instance, in the sentence '阿西娜豪华冰箱生产基地项目"正在推进过程中，预计项目一期将于2012年6月投产', annotators should label the universal phrase, '豪华冰箱(Deluxe Refrigerator)' rather than the longer and less common phrase, '阿西娜豪华冰箱(Athena Deluxe Refrigerator)'.

Table 1 provides several annotation instances from our annotations, which help us better understand the pre-defined tagging scheme.

Note that during the crowd-sourcing annotation procedure, our personnel directly annotate unsplit long documents, which are stored as *Word* documents with the *.docx* extension. The long text files that have been directly annotated can be accessed on GitHub.[12] We are aware that Transformer-based model training imposes limitations on the size of input and output. In order to meet the requirements for subsequent information retrievalb model training and prediction, we have appropriately divided the long documents into shorter document segments, shorter texts. The corresponding document segmentation results can be found on GitHub.[13]

After completing the annotation steps mentioned above, we have generated over 12,000 labelled instances. However, we should note that distractions during annotation may cause annotators to overlook some key-phrase labels, potentially affecting the quality of our dataset. To address this, we integrate all the labelled phrases and retroactively apply them to the previously annotated dataset to ensure comprehensive coverage of key-phrases with a certain level of quality. Through this back-labeling step, the dataset is enriched populated with up to 165 key-phrases.

3.3 Dataset Statistics

Following the crowd-sourcing of annotations, we randomly split our dataset into a training subset and a test subset. The training set comprises approximately 11,579 samples, while the test set contains about 610 samples, resulting in a ratio of approximately *19:1*. This splitting ratio follows the same schedule as previous works [35, 44]. The aforementioned partitioning can serve as a standard and specification for subsequent dataset usage. It is worth mentioning that although we have initially established this partitioning pattern, we have retained the provision to allow for a secondary partitioning of a validation set from the already partitioned training set [13, 24, 32]. The primary rationale behind this decision is to provide researchers with a certain degree of freedom in utilizing the data. We hope that future studies can effectively explore and leverage the trainable data within the existing training set, while ensuring the stability of the test set. The training and test sets available on our Github[14] can be further split using different sampling strategies for cross-validation.

[12] https://github.com/albert-jin/Fintech-Key-Phrase/tree/main/Fintech-Key-Phrase/annotated_docx
[13] https://github.com/albert-jin/Fintech-Key-Phrase/tree/main/Fintech-Key-Phrase/dataset_final
[14] https://github.com/albert-jin/Fintech-Key-Phrase



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{"text": "全年净增4亿户。随着芯片、智能识别、传感器、区块链、边缘计算等物联网相关新技术的迭代演进", "label": {"financial_entity": {"芯片": [[10, 11]], "智能识别": [[13, 16]], "传感器": [[18
{"text": "进一步完善了以智慧能源为核心的产业结构和服务体系", "label": {"financial_entity": {"智慧能源": [[7, 10]]}}}
{"text": "持续参与满足三代、四代核电关键阀门国产化的研发工作", "label": {"financial_entity": {"核电关键阀门": [[11, 16]]}}}
{"text": "向市场提供移动通信智能卡、身份识别卡、银行卡、社保卡等及其系统解决方案及相关终端产品", "label": {"financial_entity": {"移动通信智能卡": [[5, 11]], "身份识别卡": [[13, 17]], "银行卡"
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{"text": "低温共烧NiCuZn铁氧体材料，低温共烧MnZn铁氧体材料", "label": {"financial_entity": {"低温共烧NiCuZn铁氧体材料": [[0, 14]], "低温共烧MnZn铁氧体材料": [[16, 28]]}}}
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{"text": "川南厂区新建的固废焚烧系统、第二套RTO，振港厂区污水处理站和RTO等一批新增环保设施开始调试或运行", "label": {"financial_entity": {"RTO": [[17, 19], [31, 33]]}}}
{"text": "向高精端、大型化、军工领域回转支承市场转移", "label": {"financial_entity": {"军工": [[9, 10]]}}}
{"text": "体积较大的重型和中型消防车在逐渐狭小的城市中行进和展开变得越来越困难", "label": {"financial_entity": {"中型消防车": [[8, 12]]}}}
{"text": "子公司优瑞科积极拓展有色装备市场，自主设计及合作开发的圆盘浇铸机、双板刷洗机、单板刷洗机、阴极洗涤及剥片机组、进口不锈钢", "label": {"financial_entity": {"有色装备": [[10, 13]], "圆盘浇
{"text": "服务信息化系统相关的网络建设、信息化平台建设、信息安全系统建设、视频监控系统建设等方面的系统集成项目", "label": {"financial_entity": {"服务信息化系统": [[0, 6]], "信息化平台": [[15, 1
{"text": "9、产业城项目发展战略和未来一年的经营计划公司致力于提升公司北海银河科技园的科技孵化、商业服务、创新协同等综合能力", "label": {"financial_entity": {"产业城": [[2, 4]], "科技孵化": [[38,
{"text": "一系列行业政策深刻改变着市场格局和企业的未来发展，产业机会开始逐步向行业龙头集中", "label": {"financial_entity": {"来": [[21, 21]]}}}
{"text": "积极推进光伏电站项目的投资、建设与运营，同时在EPC工程推广与融资渠道拓展方面取得新进展", "label": {"financial_entity": {"光伏电站": [[4, 7]], "EPC工程": [[23, 27]]}}}
{"text": "主要用于血常规、生化等血液检测", "label": {"financial_entity": {"血常规": [[4, 6]], "生化": [[8, 9]]}}}
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{"text": "公司主导产品为中高端印制电路板，稳定熟练的员工有助于保障卓越的产品品质", "label": {"financial_entity": {"中高端印制电路板": [[7, 14]]}}}
{"text": "提高产品附加值。螯合钾尿素新产品投产", "label": {"financial_entity": {"螯合钾尿素": [[8, 12]]}}}
{"text": "二、主营业务分析1、概述公司主营业务包括生物制药的研发与销售、房地产开发和销售。生物制药包括注射用克林霉素磷酸酯、注射用三氯醋酸二钠氯化镁、注射用二氯醋酸二异丙", "label": {"financial_entit
{"text": "用于局域网（LAN）系统。低温高导材料，用于室外工作的综合业务数字网（ISDN）脉冲变压器", "label": {"financial_entity": {"局域网": [[2, 4]], "LAN": [[6, 8]], "低温高导材料": [[13, 1

Fig. 3. A screenshot of the training set in our proposed Fintech Key-Phrase. The test set is the same as the training set.

For instance, under the disclosed partitioning criteria, subsequent model evaluation experiments on various baselines were conducted based on our secondary partitioning of the initial partitioned training set. Specifically, we randomly sampled the same number of instances from the training set as the test set, creating a validation set for model real-time evaluation during a series of subsequent experiments. Thus, the actual training/validation/test ratio employed was *18:1:1*.

The storing format of the Fintech Key-Phrase is illustrated in Figure 3. Each line represents an independent instance for the expression-level information retrieval task. The attribute value ‘<text>’ denotes the sentence context, and the sub-attribute value ‘financial_entity’ in attribute ‘<label>’ contains the information of all annotated key-phrases, including the phrases themselves and their absolute positions. For example, the attribute value ‘城市轨道交通’: [[1, 6]], ‘新城镇建’: [[8, 11]], ‘硬件’: [[14, 15]] is the annotated results of the Chinese sentence ‘在城市轨道交通、新城镇建设等硬件基本工程方面，将有力拉动多种工业产品的内需市场’. In this example, the arrays [[1, 6]], [[8, 11]], and [[14, 15]] respectively indicate the absolute positions of the key-phrases ‘城市轨道交通’, ‘新城镇建’, and ‘硬件’ within the sentence.

Finally, we give comprehensive statistics for our Fintech Key-Phrase dataset. The training set we split contains annotations of over 35,884 fin-tech domain key-phrases, which are reduced to 11,434 unique key-phrases after removing duplicated phrases. Similarly, the test set contains more than 1,769 Fintech key-phrases, with 1,439 different key-phrases after removing duplicated phrases. As shown in Figure 4, we also collected statistics on the distribution of key-phrases in different length segments. The statistics reveal that the majority of fin-tech domain key-phrases fall within the length range of 1 to 6, with a significant number of key-phrases exceeding 7 in length. This observation suggests that the key-phrases of interest to financial experts are typically short and concise.

Furthermore, we gathered statistics on the distribution of annotated phrases across pre-defined categories: High-Tech Application (e.g., ‘空管实时指挥系统 and 配电网故障指示器定位系统’), Science Concept (e.g., ‘5G通信, 手机支付 and 污染治理工艺’), Industrial Products (e.g., ‘新型特种印制电路板 and 超细旦纤维’), Foreign Term Abbreviation (e.g., PVC, LDPE, and 1066Mbps-DDR3), and Others (e.g., ‘高端人才 and 自主知识产权’). It is important to note that certain

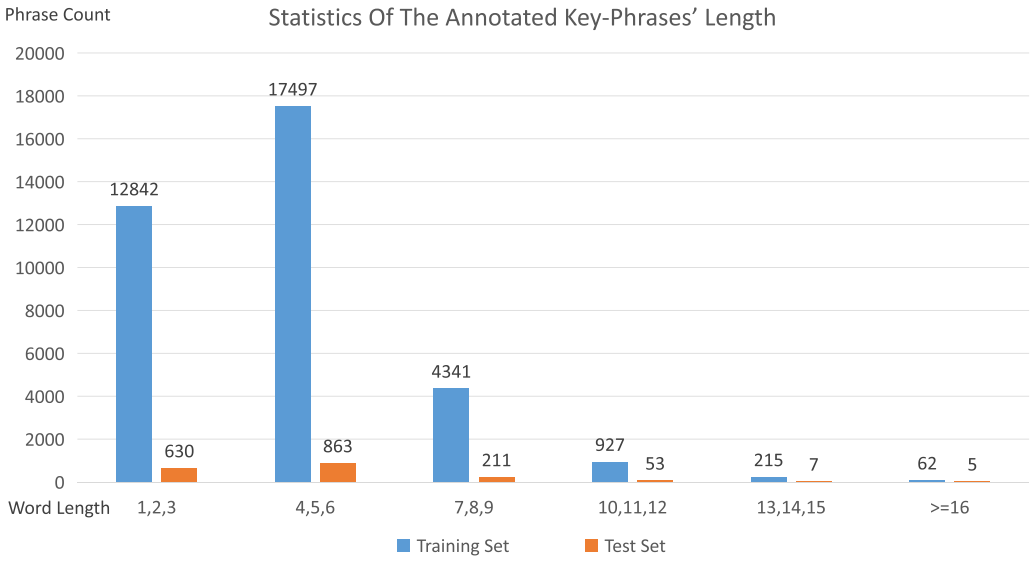


Fig. 4. The length scattering of the split train set (Blue) and the test set (Orange).

Table 2. The Financial High-tech Domain-specific Category Statistics for All Annotated Key-phrases

	High-Tech Application	Science Concept	Industrial Products	Foreign Term Abbreviation	Others
Train set	4,492	3,157	4,749	662	836
	39.29%	27.61%	41.53%	5.79%	7.31%
Test set	584	379	582	88	110
	40.58%	26.34%	40.44%	6.12%	7.65%

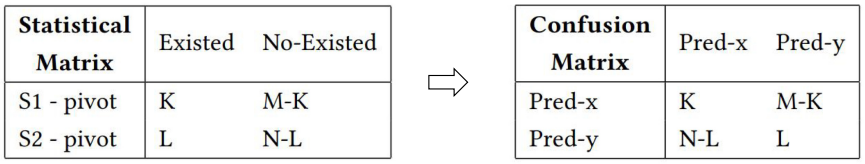
phrases may belong to multiple categories, resulting in larger summary statistics for each category compared to the original phrase count. The corresponding statistics are presented in Table 2.

3.4 Noise Evaluation

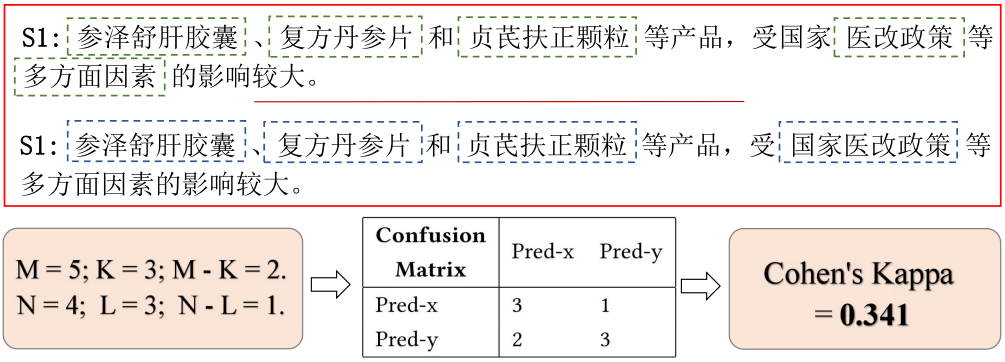
During the prolonged annotation process of the Fintech Key-Phrase dataset, different annotators may have variations in their understanding of semantic and normative aspects of annotation. Furthermore, the extended duration and high intensity of the annotation task may result in instances of incorrect annotation due to lack of concentration. These factors pose challenges to the standardization and adherence to guidelines in the proposed dataset. Therefore, in this section, we will focus on evaluating the presence of noise and mislabeled instances within the Fintech Key-Phrase dataset, to assess the quality, suitability, and limitations of the dataset.

Specifically, we conduct two approaches to evaluate the noise level in the dataset using two distinct methods: *annotator consistency assessment* and *annotation quality evaluation*.

— **Annotation consistency assessment:** The first approach involves comparing the annotations provided by different annotators and analyzing the degree of agreement among them. Through this assessment, we aimed to quantify the extent to which annotators' personal characteristics affect the consistency of Fintech Key-Phrase;
— **Annotation quality evaluation:** The second approach involves calculating the actual error rates or indicators to identify existing annotation errors. This primarily focuses on assessing the data annotation quality and quantifying the noise level of Fintech Key-Phrase.



(a.) The statistical and confusion matrices toward the annotated document S1 and S2.



(b.) The calculation example for Cohen's Kappa coefficient by an annotated instances, S1 and S2.

Fig. 5. The figure includes the definition of the confusion matrix (a) and a specific example (b) that demonstrates the calculation process of Cohen's kappa coefficient.

3.4.1 Annotation Consistency Assessment. To assess annotator consistency, we designed a specific experiment as follows. We randomly selected 100 samples from the annotated dataset, dividing them equally into training and testing sets with 50 samples each. This allocation ensured a stable training process and reliable predictions. Subsequently, we engaged the services of three different annotators who performed sentiment annotation on these samples. These annotators are all financial experts from the School of Business at Nanjing University,[15] with a solid financial background and a specialized understanding of the fin-tech field, ensuring the premise of the accurate dataset noise evaluation standards.

Moreover, the level of agreement among annotators was evaluated using the Cohen's Kappa coefficient [36], a statistical measure specifically designed to assess consistency in classification or labeling tasks. Note that the Cohen's Kappa coefficient ranges from −1 to 1. A large value that is close to 1 indicates perfect agreement, 0 indicates agreement by chance, and negative values indicate agreement worse than chance. Typically, a value above 0.5 is considered strong agreement, while values below 0.3 indicate poor agreement. Specifically, during our evaluation, the Cohen's Kappa coefficient can be regraded as follows.

Given the information retrieval results for a single text snippet S, which has been annotated by two different annotators, annotator A and annotator B. We denote the annotations provided by annotator A and annotator B as $S1$ and $S2$, respectively. Based on the above settings, the statistical matrix U (on the left) and the confusion matrix X (on the right) representing the matching degree of annotators A and B for the text snippet S can be seen in Figure 5(a). Simultaneously, the objective of the Cohen's Kappa coefficient is to assess the consistency between $S1$ and $S2$ in terms of the retrieved information from the annotated text. The calculation process is depicted in Equations (1)

[15] https://nubs.nju.edu.cn/main.htm

and (2):

$$Cohen's \quad Kappa \quad = \quad \frac{p_o - p_e}{1 - p_e}, \tag{1}$$

$$where \quad p_o = \frac{K + L}{(N - L) + (M - K)} \quad \& \quad p_e = \frac{(K + N - L) \times K + (L + M - K) \times N}{(N + M)^2}, \tag{2}$$

where p_o is the observed proportion of agreement between the two raters, and p_e is the expected proportion of agreement between the raters by chance.

Based on the aforementioned steps, we compute the Cohen's Kappa coefficient for each annotation between two different annotators. By averaging the individual Kappa coefficients, we derive the mean Cohen's Kappa coefficient, serving as a measure of consistency among the pairwise annotators. For the three hired experts paired in all possible combinations, we perform the calculation thrice, resulting in three Kappa coefficients. Subsequently, we take the average of these coefficients to obtain the overall global Cohen's Kappa coefficient for the Fintech Key-Phrase dataset. The specific results are summarized in the following table:

Table 3. The Consistency Evaluation Results of Cohen's Kappa Coefficients for the Three Experts' Annotations

Annotator Pairing	Expert 1 - Expert 2	Expert 1 - Expert 3	Expert 2 - Expert 3	**Average**
Kappa Coefficient	0.865	0.742	0.797	**0.801**

According to the consistency assessment results between the pairwise annotators calculated above as shown in Table 3, the averaged Cohen's Kappa coefficient we obtained is **0.801**, which is above **0.5**. This indicates a relatively high level of agreement among the annotators. This means that there is some slight inconsistency in the annotations within the dataset, but the level of noise associated with the annotation consistency is relatively low. After further observation of inconsistent labeling results, it was found that the main reason for a small number of inconsistencies in the annotations is due to the annotators' lack of a unified understanding of the requirement of *Suffixes Exclusion Rules*, resulting in labeling errors. For instance, Regarding the sentence '发扬光大中医药文化，充分发挥中国几千年中医药美容的传统', the ideal annotation should include '中医药文化' (traditional Chinese medicine culture) and '中医药美容' (traditional Chinese medicine cosmetology). The annotator only annotated '中医药' (traditional Chinese medicine), while overlooking the fact that '文化' (culture) and '美容' (cosmetology) are also meaningful terms in the sentence. Another example is the sentence '华帝公司隆重发布华帝魔碟灶、魔镜V2烟机和华帝魔箱智能蒸烤一体机', the ideal annotation should exclude the company-specific prefixes, such as '华帝' and '魔镜V2', as they are phrases that do not contribute significantly to the overall meaning. However, the annotator mistakenly included them, resulting in incorrect annotations.

By evaluating the annotation consistency among different annotators, we conducted a comprehensive assessment of the annotation quality of Fintech Key-Phrase. In the future, we will further improve the quality of the dataset by providing standardized training for annotators, introducing additional validation steps, expanding the dataset scale, or utilizing ChatGPT for semantic verification and revision, among other measures.

3.4.2 Annotation Quality Evaluation. In addition to evaluating annotation consistency, we also conducted an assessment of the absolute quality of the dataset's annotations, which we refer to as *annotation quality evaluation*. Unlike *annotation consistency assessment*, the former is primarily to assess the impact of annotation habits discrepancies among different annotators on Fintech Key-Phrase, while annotation quality evaluation is mainly aimed at evaluating the annotations

Table 4. The Typical Fintech Key-Phrase Annotation Noises

No.	Sentence	Type	Results
S1	储备了微生物氮磷增效、海藻提取物自产等一批新技术。	ideal	微生物氮磷,海藻提取物
		anno	微生物氮磷增效,海藻提取物
S2	磁性器材、电池、电子产品、晶体硅太阳能电池片、太阳能硅片及组件的生产、销售。	ideal	磁性器材,电池,电子产品,太阳能电池片,太阳能硅片,组件
		anno	磁性器材,电池,电子产品,晶体硅太阳能电池片,太阳能硅片及组件
S3	部分产品增加无人驾驶技术、人脸识别功能、360度环视功能和车联网技术。	ideal	无人驾驶,人脸识别,360度环视,车联网
		anno	无人驾驶,人脸识别功能,360度环视功能,车联网

The words highlighted in red indicate the incorrect annotations in comparison to the ideal labeling.

accuracy of the Fintech Key-Phrase compared to the ideal annotations, the unanimous consensus reached by experts' consultation and discussion.

In particular, based on the three different sets of annotations provided by the three annotators for 100 sample examples in the aforementioned evaluation, we highlighted the inconsistent annotations. We then invited three experts to discuss the areas of inconsistency and ultimately reached a consensus, unifying the annotations that had labeling ambiguity. Comparing this post-negotiation idealized annotation with the corresponding annotation in the Fintech Key-Phrase, we have drawn the following observations.

— Based on the sampling and statistical analysis, it was found that there were 7 erroneous sentences out of a total of 100 test samples, accounting for 7% of the total sample;
— The total number of erroneous phrases was 12, while the total number of phrases in the sampled data was 274. Therefore, the erroneous phrases comprised only 4.38% of the total annotated phrases;
— Based on the comparative observations of these noises, we have determined that the main cause of these erroneous annotations is the inconsistency in the annotators' understanding of the *Suffixes Exclusion Rules*;
— It can be concluded that despite the presence of a small amount of noise, the impact on the subsequent model's semantic understanding is minimal. The overall annotation quality of the Fintech Key-Phrase dataset is relatively high.

In order to provide readers with a clearer understanding of the dataset annotation noises, we selected several typical noise instances and presented them in Table 4.

Hence, the comprehensive evaluation has provided us with a nuanced understanding of the strengths and weaknesses of Fintech Key-Phrase, enabling its more effective utilization in serving the finance and high-tech industries. To address the challenge of annotation noise, we will explore diverse solutions to optimize Fintech Key-Phrase. These measures may encompass intensifying annotator training and leveraging the capabilities of ChatGPT to restructure the expressions of the dataset. By adopting such approaches, we aim to enhance the overall quality and usability of Fintech Key-Phrase for future applications.

4 EXPERIMENTS

To further investigate the significance of our Fintech Key-Phrase and provide explicit baseline comparisons for the expression-level information retrieval task, we conducted extensive experiments using a series of SOTA neural network-driven methods. Given the exhaustive and comprehensive nature of our experiments, we believe that the reported experimental results in this work qualify as representative baselines for future domain-specific information retrieval-related research.

In the next section, we provide detailed task definitions (Section 4.1); introduce the baselines and metrics (Sections 4.2 and 4.3); describe the detailed architectures and training procedures of the baselines (experimental analysis: Section 4.5); and present the information extraction performances of these models (performance statistics: Section 4.6).

4.1 Task Definitions

The expression-level information retrieval task can be transformed into a classic sequence tagging problem, where the model aims to assign a category to each word from a set of pre-defined labels. In our experiments, we followed the traditional 'B, I, O, S' tagging scheme. Here, the tag 'B' denotes the beginning position of any interesting key-phrase; the tag 'I' denotes the inner-position (including the end position) of any interesting key-phrase; the tag 'O' represents the position of other tokens that are not inside our attentions, and the tag 'S' signifies the position of an interesting one-word key-phrase (single token). Due to the unique characteristics of our Fintech Key-Phrase dataset, only one category 'financial_entity' is used to represent all key-phrases. However, we define four tags including 'B-financial_entity', 'I-financial_entity', 'S-financial_entity' and 'O', which are designed to extract the majority of key-phrases.

4.2 Baseline Models

Although RNNs-based methods have demonstrated their powerful ability to learn contextual representations, pre-trained word embeddings (e.g., Word2Vec [29] and GloVe [18]) alone are not sufficient to capture the contextual semantic features present in textual sentences [42]. Additionally, pre-trained embeddings are often obtained through training on small-scale corpora. The transfer of knowledge from pre-trained embedding-based RNN methods to other NLP tasks or fields typically presents several difficulties and challenges [16, 47], as extensively validated by many researchers in NLP-related studies [20, 44] through numerous comparison experiments. Hence, all embedding-based methods are excluded from our experiments. The baseline models we have chosen all rely on **pre-trained language models** (**PLMs**) such as Chinese-BERT [1, 6], Chinese-RoBERTa [27], and Chinese-ELECTRA [3].

Based on the publicly released code on Github[16] and its optimized experimental settings, we employed six information extraction models to evaluate their different performances. The main distinction among these models lies in their combination of various neural networks, such as a language encoding network layer (e.g., BERT and RoBERTa), and a label decoding network layer such as SoftMax or **conditional random fields** (**CRF**). The SOTA baseline models we adopted are as follows:

- **BERT-Linear:** This approach is a simple yet effective approach which directly adds a single label decoding classifier after the feature encoder BERT. In our experiments, we utilize a *full-connected layer* (Linear) after the BERT encoder to decode the output hidden state of each BERT token, along with a *SoftMax* operation to determine the probability of each token's label category;
- **RoBERTa-Linear:** In this model, we replace the Transformer BERT encoder of *BERT-Linear* with the enhanced Transformer known as '**A Robustly Optimized BERT**' (**RoBERTa**) to better utilize the advantages of large-scale PLMs;
- **BERT-CRF:** This BERT-based labeling model replaces the simple *Full-Connected* classifier with a more complex conditional random field [7] classifier, as compared to the *BERT-Linear* model;

[16] https://github.com/hemingkx/CLUENER2020

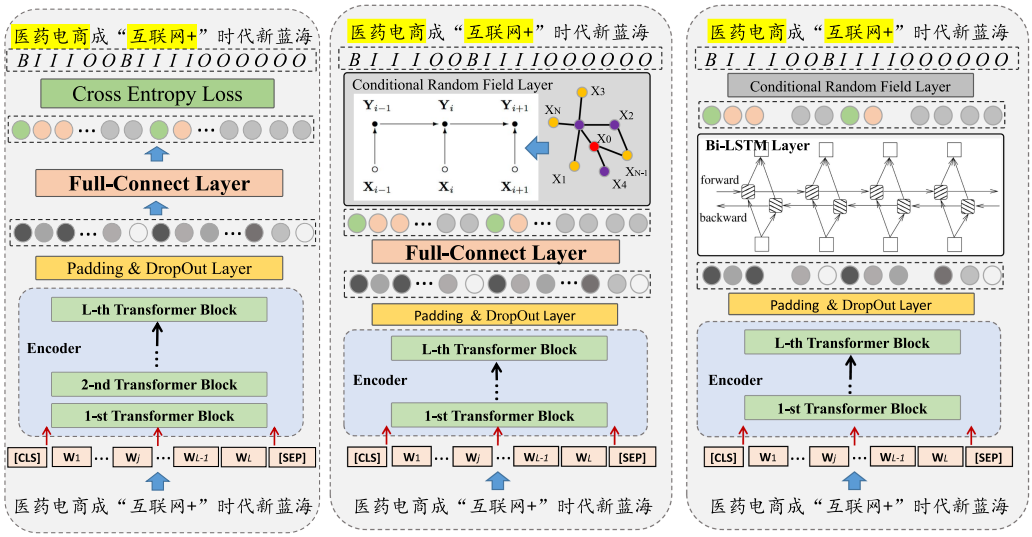


Fig. 6. The overall architectures of all the baselines we adopted. The feature encoders can be replaced with various Transformer-based representation models. The yellow highlighted expressions in language sentences represent the final extracted information of the baselines.

- **RoBERTa-CRF:** Similar to *BERT-CRF*, this model replaces the Transformer BERT encoder with the advanced RoBERTa encoder to achieve improved information retrieval performances;
- **BERT-BiLSTM-CRF:** This model introduces an extra **Bidirectional Long Short Term Memory** (**Bi-LSTM**) network between the BERT encoder and the CRF classifier to better capture dependencies in observation sequences;
- **RoBERTa-BiLSTM-CRF:** In this model, the Transformer BERT encoder from *BERT-BiLSTM-CRF* is replaced with the enhanced Transformer RoBERTa to achieve superior performance.

The inner-construction details of each baseline model's architecture will be illustrated in Section 4.5, *Experimental Analysis*.

4.3 Evaluation Metrics

Following previous works [16, 17], we adopt the F1-score metric as the standard evaluation metric, which is also commonly used in multi-label classification tasks. Tthe F1-score can be seen as a weighted balance between the *Recall* rate and *Precision* rate. It ranges from 0 to 1, with 1 indicating the best performance and 0 indicating the worst. While the *Accuracy* metric is widely used in many prediction tasks, it may not effectively reflect the overall quality of predictions due to the unbalanced proportion and distribution of the pre-defined tags, namely 'B I O S'. In contrast, the F1-score a more comprehensive assessment of the various features of these tags. Hence, we choose the F1-score as our metric for evaluation. Generally, a higher F1-score corresponds to a more powerful information extraction system. The evaluation metric F1-score is calculated using the following formula:

$$f1 = \frac{2 * Prec * Rec}{Prec + Rec} \qquad where \quad Prec = \frac{TP}{TP + FP} \quad \& \quad Rec = \frac{TP}{TP + FN}. \tag{3}$$

Where *Prec* and *Rec* represent the precision rate and recall rate of the label classification model, respectively; *TP* (True Positives) is the number of samples where the actual class is positive and the predicted class is also positive; *FP* (False Positives) is the number of samples when the actual class is negative and the predicted class is positive; *FN* (False Negatives) is the numbers of samples when the actual class is positive but the predicted class is negative.

4.4 Hyperparameter Settings

As for obtaining optimal experimental results, hyperparameter selections are particularly important. However, considering that the reliability of multiple baselines depends on their comparative consistency, we made several minute-specific adjustments for some baselines while retaining the major hyperparameter choices of others. Thus, we illustrate the universal hyperparameters here and leave the specific hyperparameters introduced in each corresponding experimental analysis part.

Our code implementation is based on Python 3.7[17] and PyTorch 1.9.0.[18] We conducted our experiments on a system running textitWindow 10 Professional equipped with a *GTX 1080Ti*. The pre-trained transformer BERT model [6] we utilized, called 'Bert-Base-Chinese',[19] was developed by Cui et al. [3] from the **Research Center for Social Computing and Information Retrieval** (**RCSCIR**) at **Harbin Institute of Technology** (**HIT**). 'Bert-Base-Chinese' consists of 12 transformer blocks, 768 hidden sizes, 12 Self-Attentions Heads, and approximately 392M parameters. For our experiments, the pre-trained transformer RoBERTa [3, 27] that we employed is 'Chinese-RoBERTa-wwm-large',[20] which is derived from the original BERT. 'Chinese-RoBERTa-wwm-large' incorporates dynamic mask strategy and is trained on a larger Chinese language corpus with a larger training mini-batch. It comprises 24 transformer blocks, 1,024 hidden sizes, 16 Self-Attentions Heads, and around 1,210M parameters. The vocabularies of Chinese-BERT and Chinese-RoBERTa are the same, containing approximately 21,128 different words.

In addition to splitting the Fintech Key-Phrase dataset into training and test sets, we randomly separate 1/20 of the instances from the train set to create a development set. This development set aided in determining when to stop the training process. The default learning rate was set to 2×10^{-5}. As a regularization method, we employed Weight Decay instead of L_2 normalization, with the Weight Decay rate of 0.01 to mitigate overfitting. To address the gradient explosion issue, we implemented gradient clipping, capping all computed gradients at 5. We set the maximum epoch count to 50 to ensure sufficient training steps for learning from the Fintech Key-Phrase dataset. Additionally, we employed the early-stopping strategy by evaluating the F1-score every 5 training epochs on the validation set to prevent overfitting. The batch size of BERT-based models was set to 32 to fully utilize the GPU's capabilities. To avoid GPU RAM memory constraints, the batch size for RoBERTa-based models was reduced to 6 for each iteration. However, the smaller batch size would resulted in decreased experimental performance. Although RoBERTa-based models did not exhibit significant superiority over BERT-based models in our experiments, we plan to leverage a more powerful GPU (such as the *RTX 3090Ti*) to achieve more competitive performances in the future. Finally, we employed the AdamW (Adam with weight decay fix) [28] stochastic gradient descent algorithm to update the optimized parameters of the model by back-propagating the computed loss.

[17] https://www.python.org/downloads/release/python-370
[18] https://pytorch.org/blog/pytorch-1.9-released
[19] https://storage.googleapis.com/bert_models/2018_11_03/chinese_L-12_H-768_A-12.zip
[20] https://github.com/ymcui/Chinese-BERT-wwm

4.5 Experimental Analysis

This section presents a comprehensive overview of the adopted baselines, including detailed information about each model, as well as the overall experimental results. Additionally, we provide a thorough analysis of the training, validation, and prediction procedures.

It is widely recognized that pre-trained word embedding techniques have certain limitations in NLP tasks. Firstly, pre-trained fixed embeddings fail to capture the rich contextual semantics present in a sentence. Secondly, RNNs process information sequentially, word by word, which hampers parallel processing efficiency. To overcome these limitations, we chose to base our baselines on the popular BERT and its variants.

4.5.1 BERT-Linear. In recent research, pre-trained Transformers have demonstrated their potential advantage in modelling long-range textual context and have significantly facilitated training and predicting time due to their parallel computing capabilities. In our experiments, we first utilized the baseline model called *BERT-Linear* to better explore the overwhelming advantages of Transformers. This model consists of a pre-trained Transformer called BERT, which serves as the feature encoder, and a simple Linear & SoftMax joint layer that acts as the tag classifier (also known as the feature decoder).

The Google AI team [6] introduced the well-known **Bidirectional Encoder Representations from Transformer** (**BERT**) in 2018. BERT is a pre-trained language representation model [1] that can be fine-tuned for a wide range of NLP tasks. Due to its widespread superiority in recent years, we will provide an exhaustive introduction to the architecture of BERT.

BERT consists of multiple identical Transformer blocks stacked together, aiming to capture capture contextual semantic features of linguistic sequences more effectively. By pre-training on large-scale corpora, BERT has acquired substantial prior knowledge, which greatly benefits numerous **natural language understanding** (**NLU**) tasks. Leveraging this extensive prior knowledge, BERT achieves SOTA performance in various downstream NLP tasks without requiring significant task-specific architectural modifications [20, 34, 35].

Following the fine-tuning settings of Transformers, we tokenize the original sentence *Sent* (e.g., '医药电商成互联网时代新蓝海') using the WordPiece tokenizer.[21] The resulting separated tokens are then combined into an integrated token sequence of length L, denoted as $S = \{W_i\}_1^L$. The predefined special tokens '[CLS]' and '[SEP]', respectively, indicate the beginning and end positions of a language sentence. These tokens are used to encapsulate the input sentence S into a single sequence: $[CLS]; S; [SEP]$. We refer to the concatenated token sequence $\{[CLS]; S; [SEP]\}$ as $T = \{t_j\}_{j=1}^{L+2}$. For each token t_j at index j of the concatenated sequence T, a corresponding positional embedding $p_j (j \in \{1, \ldots L+2\})$ is concatenated with it to create the initial token embedding $s_j (j \in \{1, \ldots L+2\})$. These embeddings are then fed into the Transformer blocks. Given an original sentence, denoted as *Sent*, the corresponding formulas are presented as the following equations.

$$S = \{W_i\}_1^L = [W_1; W_2; \ldots; W_L] = WordPiece(Sent), \tag{4}$$

$$T = \{t_j\}_{j=1}^{L+2} = \{[CLS]; S; [SEP]\} \qquad where \qquad S = \{W_i\}_1^L, \tag{5}$$

$$S = \{s_j\}_1^{L+2} = CONCAT(T; P) = CONCAT(\{t_j\}_1^{L+2}; \{p_j\}_1^{L+2}) \qquad where \qquad s_j = concat[t_j; p_j]. \tag{6}$$

After the pre-processing procedure, the initial representation of the entire sentence, denoted as $S = \{s_j\}_1^{L+2}$, is passed through successive Transformer encoder blocks. The Transformer blocks

[21] https://towardsdatascience.com/word-subword-and-character-based-tokenization-know-the-difference-ea0976b64e17

are denoted as $\{TransBlock_k\}_{k=1}^{N}$, where $TransBlock_k$ represents the kth layer of BERT; The depth of BERT and the number of Transformer blocks is denoted by N ($N \in \{8, 12, 16, \ldots\}$). The Transformer encoder generates the hidden representation h_j^N for each input token S_j^0, which serves as the output of BERT. The final context-aware representation for the input sequence is denoted as $H^N = \{h_j^N\}_{j=1}^{L+2}$. The corresponding formulas are in Equations (7) and (8):

$$h_j^N = Transformer(h_j^0) = \{TransBlock_k\}_{k=1}^{N}(h_j^0), \tag{7}$$

$$H^N = Transformer^N(H^0) = \{TransBlock_k\}_{k=1}^{N}(H^0)$$
$$where \qquad H^i = \{h_j^i\}_{j=1}^{L+2} = [h_{[cls]}^i, h_1^i, h_2^i, \ldots h_L^i, h_{[sep]}^i]. \tag{8}$$

Where H^i represents the feature representations of the $i-th$ layer of BERT, and $\{h_j^i\}_{j=1}^{L+2}$ denotes the representation sequence of each token at the $i-th$ Transformer block, including the special tokens '[CLS]' and '[SEP]'.

Afterwards, as shown in Equation (9), the acquired hidden state sequence H^N undergoes a padding operation to obtain a new state sequence with a standardized length $\widehat{H}$. Subsequently, a dropout layer is applied to randomly drop a portion of weights from $\widehat{H}$ to maintain robustness during model training. Then, as depicted in Equation (10), a full-connected layer with bias, along with a *SoftMax* operation acting as the feature decoder, is constructed to classify the final tags for all tokens in the input sentence based on the aligned states $\widetilde{H}$. After these steps, the hidden state sequence H^N outputted from the BERT module is fed and transformed into a sequence of probability logits $\{P^j(\theta)\}_1^L$ for all tag categories. Here, $P^j(\theta)$ means the predicted probability distributions for the $j-index$ token, also denoted as $\{p^j(\theta)\}_{i=1}^{T}$, where T denotes the pre-defined count of label categories. The value $p_i^j(\theta)$, ranging between 0 and 1, represents the probability of a certain token at the $j-index$ position towards the $i-th$ label type. To evaluate the similarities between the predicted tags and their corresponding gold (real) tags, we employ the commonly used *Cross Entropy* [5] as the loss function.

$$\widetilde{H} = \{\widetilde{h}_i\}_{i=1}^{T} = Dropout(\widehat{H}) \qquad \& \qquad \widehat{H} = Padding(H^N), \tag{9}$$

$$Loss(Y|Sent, \theta) = CrossEntropy(Y, \{P^j(\theta)\}_{j=1}^{L}) \qquad \& \qquad P^j(\theta) = \{p_i^j(\theta)\}_{i=1}^{T} = SoftMax(W^\top(\widetilde{h}_j) + b). \tag{10}$$

Where H^N represents the output states provided by the last layer of the BERT encoder; N denotes the number of BERT blocks; $\widetilde{h}_i$ refers to the hidden state of the $i-th$ token after the padding and dropout operations; $W^\top$ and b are the weight matrix and bias of the Linear layer, with $W^\top$ having an input dimension of 768 (BERT's hidden layer dimension) and an output dimension equal to the count of pre-defined tags; $p_i^j(\theta)$ represents the probability of the jth token towards the ith tag; $P^j(\theta)$ denotes the predicted probability distributions of the $j-index$ token in a sentence; and θ refers to all optimizable network parameters, including BERT fine-tuning parameters, $W^\top$ and b. Y is the target tag sequence (e.g., [B, I, O, O, S, . . .]). For each instance, including a sentence $Sent$, a target annotation Y, and a model with parameter θ, the loss function $Loss(Y|Sent, \theta)$ calculates the final model prediction loss using the *Cross Entropy Loss* function.

Due to the lightweight nature of the adopted BERT model, specifically 'Bert-Base-Chinese', which has relatively fewer parameters, it can be trained with a larger batch size. Therefore, in our experiments, the batch size for all BERT-based baselines is set to 32. Taking a complete training procedure as an example, the optimization processes with F1-score performances on the validation set are illustrated in Figure 7. From the line chart, it is evident that the *BERT-Linear* model converges in just 4 epochs. The rapid convergence can be attributed to the larger batch size and the

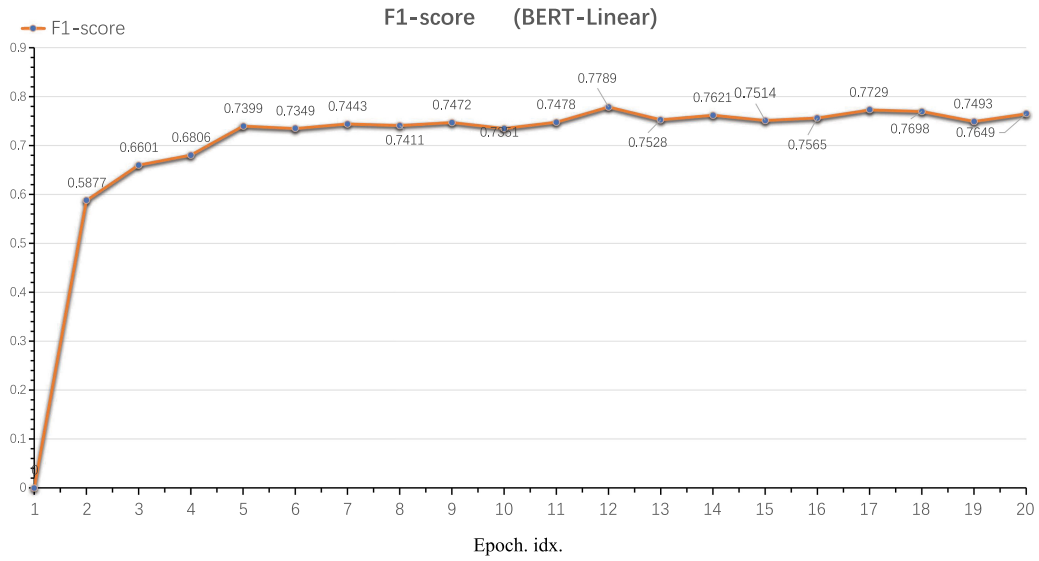


Fig. 7. The F1-score indicator evaluation statistics on the Validation dataset for each training epoch throughout the entire model optimizing process of the BERT-Linear baseline.

efficient fine-tuning optimization of BERT. Additionally, the high quality of our annotated dataset, *Fintech Key-Phrase*, which contains minimal external noise information, also plays a crucial role.

As the model approaches convergence, the F1-score fluctuates within the range of 0.75 to 0.77+. Notably, the model achieves competitive performances, reaching up to 77% in the 12-th epoch and the 17-th epoch. After obtaining the best F1-score metric (77.89%) in the 12-th epoch, the training process is halted in advance since the performance did not improve for five consecutive epochs (as determined by the pre-set hyperparameter: patience epochs). Subsequently, a temporary checkpoint model is stored and evaluated on the test set to quantitatively assess the predicting superiority of *BERT-Linear*.

It is important to note that random fluctuations during experimentation can significantly impact the final model performance. These unexpected factors may include the random seeds of CPU or GPU, which are used to initialize model parameters (e.g., Linear Layer and Dropout Layer), as well as minor differences of hyperparameter settings. To ensure the experimental robustness and stability, we perform ten-fold Information Extraction cross-validation instead of conducting a single training and evaluation cycle on our Fintech Key-Phrase dataset. Ultimately, as demonstrated in Section 4.6, the average value of all experimental F1-score performances serves as the baseline performance of *BERT-Linear* for the expression-level information retrieval task.

4.5.2 RoBERTa-Linear. To fully explore the capabilities of Transformer-based PLMs, we conducted experiments using a variant of the Transformer model, called *RoBERTa* [27] (**R**obustly **o**ptimized **BERT** **a**pproach). Similar to *BERT-Linear*, our baseline model *RoBERTa-Linear* utilizes the same tag decoder (Linear with SoftMax classifier), but replaces the feature encoder *Chinese-BERT* with the more powerful *Chinese-RoBERTa-wwm-large*.

Compared with the original *Chinese-BERT*, *Chinese-RoBERTa-wwm-large* offers several enhancements, which are as follows:

— In terms of architecture configurations, *Chinese-RoBERTa-wwm-large* follows the settings of the '*BERT-large*' (with 16 Self-Attentions Heads, Hidden Size of 1,024, and 24 Transformer

Blocks). The parameter volumes of our adopted *Chinese-RoBERTa-wwm-large* is much larger than that of our adopted *BERT*, exceeding 1GB parameters.

— *RoBERTa* optimized the **next sentence prediction** (**NSP**) task compared to the feature of original *BERT*. Several alternative training formats were proposed to investigate the importance of the NSP loss. *RoBERTa* adopts the *FULL-SENTENCES* NSP strategy, removing the next sentence prediction. This strategy ensures semantic continuity between document boundaries by continuously sampling input sentences from multiple documents and adding an extra separator token between them.

— The original BERT employs a single random masking during data processing (referred to as 'Static Masking'). However, for pre-training over more steps on large-scale datasets, this approach becomes inefficient and impractical. In contrast, *Chinese-RoBERTa-wwm-large* uses *Dynamic Masking*, performing random masking for each training instance in every epoch. This strategy significantly expands the data diversity by duplicating each training language sequence multiple times with different masking patterns, thereby efficiently utilizing the training data;

— BERT uses a WordPiece tokenizer that may split complete words into small and fragmented pieces, which can be problematic. To effectively mitigate this issue, *RoBERTa* introduced a novel technique called **Whole Word Masking** (***WWM***). With the adoption of *WWM*, the new WordPiece tokenizer, named **Chinese Word Segmentation** (**CWS**) tool, avoids excessive splitting of the text into smaller fragments. As a result, Chinese characters, which are not formed by alphabet-like symbols (individual Chinese characters), are split more accurately. This improvement of *WWM* helps the model better understand the semantics of the input text better, leading to improved performance across various NLP tasks.

— *RoBERTa* enhances the character-level **Byte-Pair Encoding** (**BPE**) used in BERT with a hybrid encoding scheme that combines character-level and word-level representations. This enhancement enables the model to handle large vocabularies commonly found in larger and more diverse natural language corpora.

— *Chinese-RoBERTa-wwm-large* is pre-trained for more training steps using longer sequences with larger training batches, with access to more data. The *RoBERTa* model was pre-trained using 1024 V100 GPUs for approximately one day. The use of large training batches and extensive training corpora improves the perplexity of the Transformer-based encoder for the downstream language modelling tasks.

Due to the large parameter volume of *RoBERTa*, the machine is prone to encountering the ***Out-Of-Memory*** (**OOM**) exception due to limited GPU compute capability (GTX1080Ti with 11G RAM allocation). To mitigate this issue, we reduce the training batch size to 6 for each iteration. Figure 8 vividly illustrates the increasing F1-score performance during the complete training procedure of the baseline model, *RoBERTa-Linear*. The experimental results indicate that *RoBERTa-Linear* requires more training epochs to approach convergence compared to the training records of *BERT-Linear*, which achieved a competitive F1-score of 78.39% on the validation set in the 6-th epoch. The main reasons for this may be: (1) The larger *RoBERTa* model with more parameters is more challenging to optimize; (2) The smaller batch size introduces instability in the convergence process due to random fluctuations in the fed instances. After a certain number of epochs, the F1-score fluctuates between 0.78 and 0.80+. Additionally, the model consistently achieves competitive performances, reaching up to 0.80 in the 8-th epoch and the 18-th epoch. Upon reaching the best F1-score metric (80.72%) in the 18-th epoch, the training process is halted in advance as the performance does not improve over 5 consecutive epochs (pre-set hyperparameter: patience epochs). Subsequently, the temporary checkpoint model is stored and quantitatively evaluated on the test set to validate the predicting superiority of the baseline *RoBERTa-Linear*.

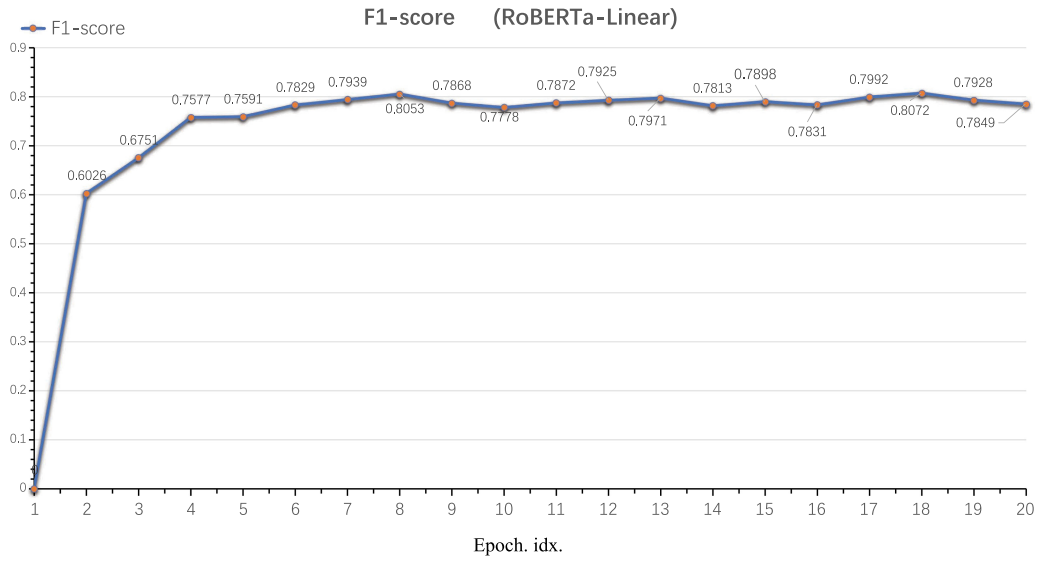


Fig. 8. The evaluation statistics of the F1-score indicator on the Validation dataset for each training epoch throughout the entire model optimization process of the RoBERTa-Linear baseline.

The statistics demonstrate that *RoBERTa-Linear* consistently outperforms *BERT-Linear* by a significant margin. Furthermore, the ten-fold cross-validation and average F1-score results of *RoBERTa-Linear* on our Fintech Key-Phrase dataset serve as representative indicators of the final performance of the baseline model. Further details are provided in Section 4.6.

4.5.3 BERT-CRF. In order to leverage the transition inter-dependency among different tags, we replace the tag classifier of *BERT-Linear* with ***conditional random fields (CRF)*** [22]. Compared with the Linear & SoftMax layer-based decoder that directly predicts the final tags, the feature decoder of *CRF* focuses on the semantic context level instead of individual position information. Unlike HMMs, *CRF* is a probabilistic graphical model that addresses the issue of abundant non-independent features [22]. Thus, the tag-related features can be better captured with this layer. Specifically, the *CRF* layer efficiently predicts the current tag by leveraging the features of past tags and future tags, as well as the distribution and transmission patterns of tags. A standard *CRF* layer includes a hidden state transition matrix that calculates the transition score from one state to another.

As depicted in the top centre of Figure 6, the output hidden states of the Transformer encoder, after a series of transforms, are fed into the CRF classifier to determine the final tag categories. The *CRF* layer constructs the transition patterns of the pre-defined tags: 'B-financial_entity', 'I-financial_entity', 'S-financial_entity' and 'O'. Given an N-length hidden state sequence $\{X_i\}_{i=1}^{N}$, the CRF algorithm calculates the probability score (rationality) between the input state sequence and the corresponding golden label sequence. Finally, the *CRF* layer provides the path with the maximum probability using the Viterbi Algorithm [9]. The following Equations (11) and (12) illustrate the *CRF* decoding operation:

$$P_w(X \mid Y) = \frac{1}{Z_w(Y)} \prod_{i=1}^{N} M_i \left(X_{i-1}, X_i \mid Y\right), \tag{11}$$

$$Z_w(Y) = \left(M_1(Y) M_2(Y) \cdots M_N(Y)\right)_{\text{start, stop}}, \tag{12}$$

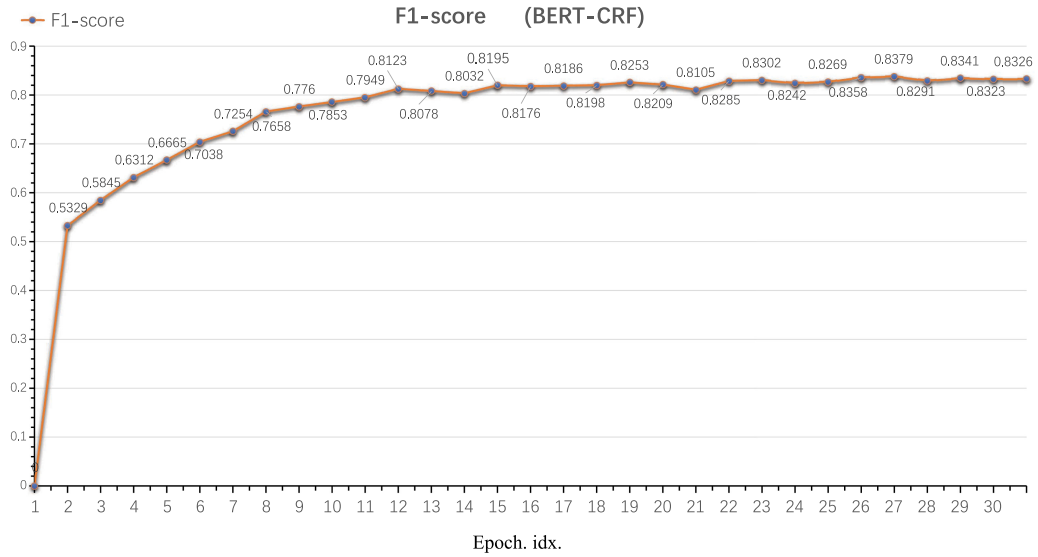


Fig. 9. The F1-score evaluation statistics on the Validation dataset for each training epoch during the entire model optimization process of the BERT-CRF baseline.

where w denotes the parameters stored in the *CRF* layer; $P_w(X \mid Y)$ represents the probability of the hidden state sequence X given the observed sequence Y; $\{X_i\}_{i=1}^N$ represents the hidden variables, and $\{Y_i\}_{i=1}^N$ denotes the observed variables. $Z_w(Y)$ indicates the degree of rationality of the observed sequence Y.

Furthermore, Figure 9 extrapolates the performance optimization process by evaluating the F1-score on Validation set during each epoch of the complete training procedure. The following observations can be made: (1) In comparison to the training records of *BERT-Linear*, *BERT-CRF* requires a significantly larger number of training epochs (up to 12 epochs) to achieve final convergence. This can be attributed to the time-consuming optimization of *CRF*, which involves complex logical patterns and a larger number of parameters; (2) *BERT-CRF* initially achieves relatively good performance (over 83% F1-score) after approximately 22 training epochs and exhibits competitive performance with a peak F1-score of 83.79% on the 27-th training epoch. The improvement in performance (an approximate 5.9% absolute F1-score gain) compared to the baseline *BERT-Linear*, which lacks *CRF*, is significant. This result validates the importance of modeling the transition patterns between tags.

4.5.4 RoBERTa-CRF. In the enhanced *RoBERTa-CRF* baseline, we replace the feature encoder Transformer-*Chinese-BERT* of the baseline *BERT-CRF* with the improved *Chinese-RoBERTa-wwm-large*, resulting in *RoBERTa-CRF*. The distinct differences between these two encoders are described in Section 4.5.2. The optimization records of the F1-score performance throughout a complete training experiment are illustrated in Figure 10.

From the line chart, it is evident that the baseline *RoBERTa-CRF* requires only around 10 epochs to reach close to final model convergence. This can be attributed to the time-consuming optimization process of CRF, which involves complex logical patterns and a larger number of parameters, requiring more training steps for full optimization. At the 23rd training epoch, the *RoBERTa-CRF* achieves the best F1-score performance of 84.07% on the validation set. Compared to *BERT-CRF*, the *RoBERTa-CRF* demonstrates a slight improvement with a 0.28% higher F1-score. It is worth noting

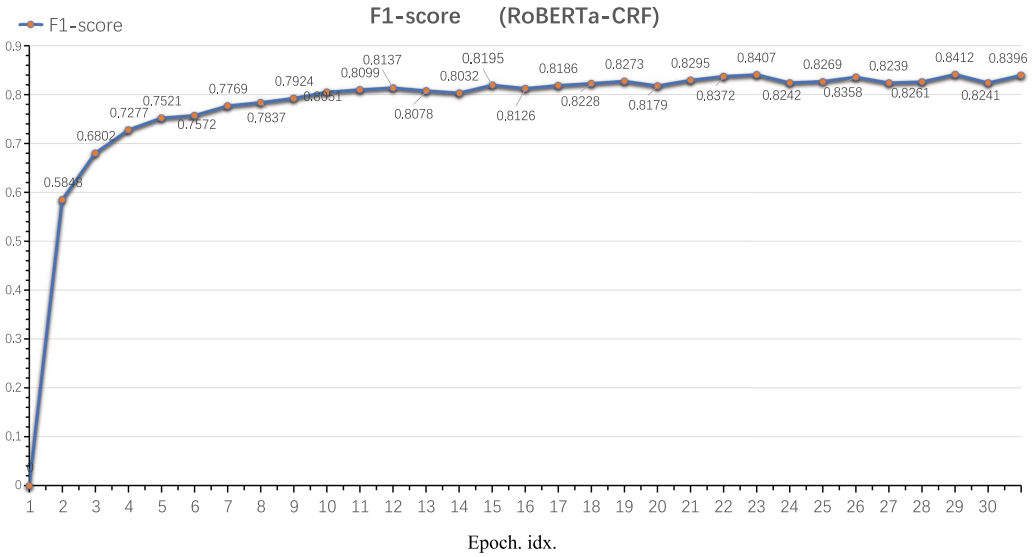


Fig. 10. The statistics of the F1-score indicator evaluation on the Validation dataset for each training epoch during the entire model optimization process of the RoBERTa-CRF baseline.

that the batch size is a crucial hyperparameter that significantly affects both training-time stability and final performance. In our experiments with RoBERTa-based baselines, we intentionally decreased the batch size from 32 to 6 to avoid potential OOM issues. Therefore, it is not surprising to observe this phenomenon. We believe that increasing the batch size would lead to more competitive performance, which we plan to investigate in future work.

4.5.5 BERT-BiLSTM-CRF. As depicted in the top-right of Figure 6, we evaluate the baseline model, *BERT-BiLSTM-CRF*, which is a well-known network architecture. In this model, an additional network module called Bi-LSTM [16] is introduced into *BERT-CRF* between the Transformer-BERT encoder and the CRF decoder as an intermediate layer. LSTM is a variant of **Recurrent Neural Networks** (**RNNs**) designed to address the issues of gradient explosion and gradient vanishing in vanilla RNNs, especially for long-term language sequence modeling. Since a single LSTM only encodes token sequence semantic features from one direction (forward or backward), it is not sufficient to capture the global contextual information. To overcome this limitation, we employ Bidirectional LSTMs, which utilize two independent LSTMs to capture features from both directions: forward and backward.

The series of Transformer-based BiLSTM-CRF models represent competitive baselines for the expression-level information retrieval task. Previous studies [20, 47] have demonstrated that the *BERT-BiLSTM-CRF* model exhibits better learning capability, resulting in improved performance compared to other models. We hypothesize several main reasons for this. Firstly, although the Transformer incorporates an extra positional embedding, the **Sin**usoidal Positional Encoding, concatenated with the original token embedding before being input to the Transformer, the bare Transformer still inevitably weakens the positional feature information among words when computing the attention feature using the *Multi-Head Self-Attention* module. Consequently, if the CRF directly decodes the Transformer's outputs, the model's ability to learn sequence-level information might be compromised. Additionally, this limitation could be further exacerbated by the deeper layers of the Transformer block. Secondly, the LSTM effectively addresses these limitations by modeling

the state sequence across all time steps. With the ability to incorporate a memory cell to retain past information over a longer period, the LSTM provides a state sequence with richer positional features compared to the direct state sequence generated by the Transformer.

Given an L-length state sequence output from BERT, the Bi-LSTM network generates both a forward hidden state sequence: $(\overrightarrow{h_{[START]}}, \overrightarrow{h_1}, \overrightarrow{h_2}, \ldots, \overrightarrow{h_L})$, and a backward hidden state sequence: $(\overleftarrow{h_{[START]}}, \overleftarrow{h_1}, \overleftarrow{h_2}, \ldots, \overleftarrow{h_L})$. The final output of the Bi-LSTM is obtained by concatenating the forward and backward hidden states for each token. The feature output of the *Bi-LSTM* is computed using the following equations, Equations (13)–(17):

$$\mathbf{f}_j = \sigma\left(\mathbf{W}_{xf}\mathbf{x}_j + \mathbf{W}_{hf}\overrightarrow{\mathbf{h}_{j-1}} + \mathbf{b}_f\right), \tag{13}$$

$$\mathbf{i}_j = \sigma\left(\mathbf{W}_{xi}\mathbf{x}_j + \mathbf{W}_{hi}\overrightarrow{\mathbf{h}_{j-1}} + \mathbf{b}_i\right), \tag{14}$$

$$\mathbf{o}_j = \sigma\left(\mathbf{W}_{xo}\mathbf{x}_j + \mathbf{W}_{ho}\overrightarrow{\mathbf{h}_{j-1}} + \mathbf{b}_o\right), \tag{15}$$

$$\mathbf{c}_j = \mathbf{f}_j \circ \mathbf{c}_{j-1} + \mathbf{i}_j \tanh\left(\mathbf{W}_{xc}\mathbf{x}_j + \mathbf{W}_{hc}\overrightarrow{\mathbf{h}}_{j-1} + \mathbf{b}_c\right), \tag{16}$$

$$\overrightarrow{\mathbf{h}_j} = \mathbf{o}_j \circ \tanh\left(\mathbf{c}_j\right), \tag{17}$$

where the variables f_j represents the activation vector of the input gate, i_j represents the activation vectors of forget gate, and o_j denotes the activation vector of the output gate. These three gate representations, f_j, i_j and o_j, are generated through multiple intricate computations. The updated memory of the LSTM is obtained by multiplying the input token feature x_j with the updating matrices W_{xf}, W_{xi} and W_{xo}. The previous old memory of the LSTM is obtained by multiplying the last hidden state $\overrightarrow{\mathbf{h}}_{j-1}$ with the forgotten matrices W_{hf}, W_{hi} and W_{ho}. $\circ$ represents the cell hidden state, σ denotes the sigmoid function, and $tanh$ denotes the hyperbolic tangent function. The logical value is then converted into a probability value between 0 and 1 using the activation operation σ.

After the semantic feature flow through the Bi-LSTM, the forward hidden state $\overrightarrow{\mathbf{h}_j}$ and the backward hidden state $\overleftarrow{\mathbf{h}_j}$ are concatenated. This yields the combined features $\mathbf{H} = \{h_{[START]}, h_1 \ldots h_j \ldots h_L\}$, where the $j-th$ hidden state $\mathbf{h}_j = [\overrightarrow{\mathbf{h}_j}; \overleftarrow{\mathbf{h}_j}]$. Finally, the concatenated states $\mathbf{H}$ are fed into the subsequent CRF classifier layer to determine the final tag categories.

Furthermore, in our specific implementation, we use half the dimension size of the Transformer encoder's hidden state (384 for BERT-based baselines, 512 for RoBERTa-based baselines) as the dimension of the hidden states in the *Bi-LSTM* module, including the hidden representations h_t and memory cells c_t. The initialization of LSTMs plays a crucial role in their performance and facilitates model convergence. Therefore, we initialize the weights of our bidirectional LSTM using Xavier initialization [14], which is significantly superior to other methods such as Gaussian and Kaiming initialization [15].

As shown in Figure 11, we observe that *BERT-BiLSTM-CRF* requires more than 15 epochs to approach final model convergence. During this phase, the F1-score fluctuates in the range of 0.82 to 0.83+. The best performance of 82.95% is achieved in the 33rd epoch. Compared to the *BERT-CRF* baseline, the performance of *BERT-BiLSTM-CRF* may be inferior in this training procedure, indicating that incorporating the Bi-LSTM layer is not always beneficial enough. Furthermore, we conducted several empirical experiments with *BERT-BiLSTM-CRF* under the same settings. Some of these experimental results outperformed *BERT-CRF* but only showed slight performance gains

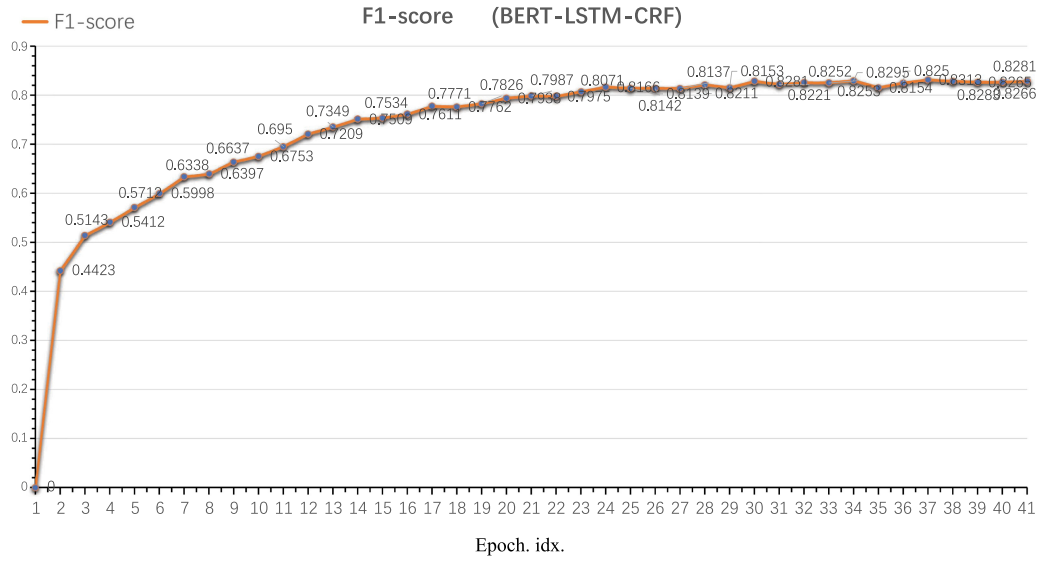


Fig. 11. Statistics of F1-score evaluation on the Validation dataset of each training epoch during the whole model optimizing procedure of the BERT-LSTM-CRF baseline.

(around 0.2%~0.7% absolute F1-score). We speculate that this is because the deeper model stacking strategy sometimes masks the semantic feature capturing capability of the *Bi-LSTM* module. In most cases, adopting BERT as a contextual feature encoder together with a simple tag decoder (e.g., Tensor-Level Operations including Linear& SoftMax layer) is relatively competitive compared to other mainstream methods. Although stacking multiple layers or adding more feature forwarding operations can bring slight performance gains, increasing the network depth sometimes leads to a significant reduction in performance.

4.5.6 RoBERTa-BiLSTM-CRF. In the baseline *RoBERTa-BiLSTM-CRF*, we replace the feature encoder Transformer-*Chinese-BERT* of the *BERT-BiLSTM-CRF* baseline with an enhanced encoder called *Chinese-RoBERTa-wwm-large*, resulting in the *RoBERTa-BiLSTM-CRF* model. The distinct differences between the two encoders were discussed in Section 4.5.2. Following a similar training pattern, the optimization records of the F1-score performance for a complete training experiment are shown in Figure 12.

From the line chart, we observe that *RoBERTa-BiLSTM-CRF* takes approximately 12 epochs to approach convergence. It achieves the best F1-score performance of 83.79% on the validation set during the 27-th training epoch. Compared to the *BERT-BiLSTM-CRF*, *RoBERTa-BiLSTM-CRF* shows a slight performance improvement with a best F1-score approximately 0.84% higher. The batch size is a critical hyperparameter that can impact training stability and final performance. It's worth noting that we significantly reduced the batch size (from 32 to 6) in all experiments involving RoBERTa-based baselines to mitigate the OOM issue. We believe that increasing the batch size would lead to a more competitive performance (up to 2% absolute F1-score gain), which is an avenue we plan to explore in future work.

4.6 Performance Statistics

It is important to emphasize that, regarding the dataset usage, the comprehensive evaluation of model metrics is based on a secondary partitioning from the training set, which serves as a reliable

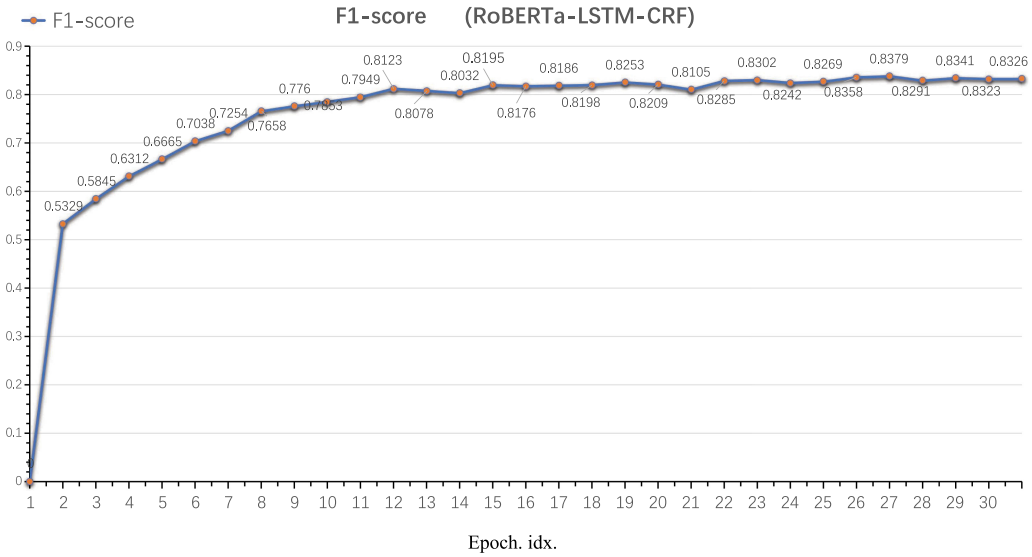


Fig. 12. The statistics of F1-score indicator evaluation on the Validation dataset for each training epoch during the entire optimization process of the RoBERTa-LSTM-CRF baseline.

Table 5. The Performance Statistics of the Ten-fold Evaluation Experiments (Under Different Random Seeds) on the *Fintech Key-Phrase* Dataset for the Six Adopted Baselines

Baseline/Experiments	1	2	3	4	5	6	7	8	9	10	Min	Max	(Avg.±Std.)
BERT-Linear	75.45	76.29	**77.92**	77.19	76.94	77.32	75.72	77.47	75.29	76.38	75.29	77.92	(76.596±0.864)
RoBERTa-Linear	79.44	**80.57**	78.37	80.16	78.92	78.79	78.37	80.16	78.71	79.32	78.37	80.57	(79.278±0.752)
BERT-CRF	81.25	80.22	80.78	**81.51**	80.31	79.92	81.41	80.75	80.49	79.41	79.41	81.51	(80.645±0.689)
RoBERTa-CRF	81.93	80.57	81.77	**82.31**	81.93	80.57	81.77	81.65	**82.31**	80.96	80.57	82.31	(81.475±0.66)
BERT-BiLSTM-CRF	80.96	81.62	81.24	80.27	81.48	81.24	80.96	**82.07**	81.60	80.48	80.96	82.07	(81.202±0.526)
RoBERTa-BiLSTM-CRF	81.77	81.62	**82.31**	80.75	80.96	81.25	81.93	81.41	**82.31**	80.75	80.75	82.31	(81.512±0.551)

The columns '*Min*' and '*Max*', respectively, indicate the minimum and maximum F1-score achieved. The column '*Avg.±Std.*' represents the average performances within the standard deviation range of all the baselines.

and accurate indicator of the model's real-time convergence throughout the entire training process. Consistent with the discussed data usage standards, our validation set comprises randomly selected examples from the training set, with the same number as the test set. As we are aware, the "Validation" set, also known as the "Development" set, is employed to assess the model's performance at each training epoch and determine the appropriate point for early stopping the training process. In contrast, the *Test* set is employed to assess the final performance by evaluating the checkpoint model that achieves the best results on the *Validation* set. Therefore, the experimental results on the test set accurately represent the ultimate capabilities of these baselines for the expression-level information retrieval task.

Furthermore, the above experimental analyses are based on a single experiment for each baseline, leading to certain random fluctuations. To mitigate the impact of various random factors, we conducted the experiments at least ten times and reported the average F1-score and standard deviation on the test set. We hope that these extensive experimental results can serve as standard baseline performances for research in the Chinese financial high technology field, specifically for the expression-level information retrieval task. We summarize the results and present these in Table 5.

From the overall F1-score performance comparisons on our *Fintech Key-Phrase* dataset in Table 5, we can clearly observe the following capability rankings of the baselines: *RoBERTa-BiLSTM-CRF* >*BERT-BiLSTM-CRF* >*RoBERTa-CRF* >*BERT-CRF* >*RoBERTa-Linear* >*BERT-Linear*. The table also highlights that the small batch size greatly limits the power of the enhanced Transformer *RoBERTa*. Furthermore, the performance gain of *BERT-BiLSTM-CRF* compared to the *BERT-CRF* (similar with *RoBERTa-CRF* and *RoBERTa-CRF* counterparts) is trivial. We speculate that this is due to the fact that stacking multiple layers may bring a slight performance gain, but sometimes the performance is reduced when the network depth increases. Disregarding these minor factors, these results further demonstrate the powerful capabilities of Transformer-based encoders, such as BERT and its variant RoBERTa, the superiority of the CRF layer over other simple classifiers, and the effectiveness of *BiLSTM* for long-term language sequence modelling. Based on our empirical explorations on the *Fintech Key-Phrase* dataset, we anticipate the emergence of a wide range of powerful SOTA methods for the expression-level information retrieval task in the future.

5 CHATGPT-BASED DATA AUGMENT FOR FINTECH-KEY-PHRASE

As we all know, the information retrieval ability of extraction models depend on various attributes of the corresponding datasets, especially the quality and dataset scale of supervised annotations. Currently, the Fintech-Key-Phrase dataset has surpassed 10,000 samples, but there is still significant room for improvement in terms of its effectiveness. To address this issue, we can enhance the performance of the information extraction model at the data level. We attempt to employ text augmentation strategies to further expand the dataset's scale.

Recently, the ChatGPTs family [45] (e.g., GPT-3.5 and GPT-4) has exhibited remarkable language generation abilities and brought extraordinary success in various cross-disciplinary NLP researches and applications, such as Event Extraction [11] and Multilingual Translation [19]. ChatGPT is a leading conversational language model developed by OpenAI, which serves as an expert in all fields with omnipotent and omniscient knowledge. Also, the ChatGPT office has provided a user-friendly interface that grants the general public unprecedented access to the capabilities of large language models.[22]

Inspired by ChatGPT's superior text processing and generalization capabilities [39, 45], we hope to utilize ChatGPT to generate augmented texts that simulate real-world semantic contexts. In this work, we explore a novel data augmentation approach based on ChatGPT, aiming to augment the Fintech-Key-Phrase dataset while improving the model's performance on expression-level information retrieval tasks. A notable example of similar work is the research conducted by Dai et al. [4], where they used ChatGPT to augment the text for each category in text classification, resulting in improved accuracy for the task. Drawing inspiration from Dai et al. s' work, we need to engage in high-quality interactions with ChatGPT to obtain effective augmented texts it can provide. We present the entire process of the text augmentation strategy in Figure 13.

Specifically, we construct a series of question-answer templates to guide ChatGPT in generating ideal responses at first. Next, we combine each annotated original text and the labeled phrases with prompting templates to form complete queries, and then send requests to ChatGPT. As depicted in Figure 13, ChatGPT will provide relevant responses containing the augmented text. Then, we further post-process the intermediate answers generated by ChatGPT to obtain the final results that can be used as an expansion for Fintech key phrases. Finally, through a series of systematic baseline model performance evaluations, we observed that the Fintech Key-Phrase processed using

[22]You can access ChatGPT by visiting the following URL: https://chat.openai.com/chat

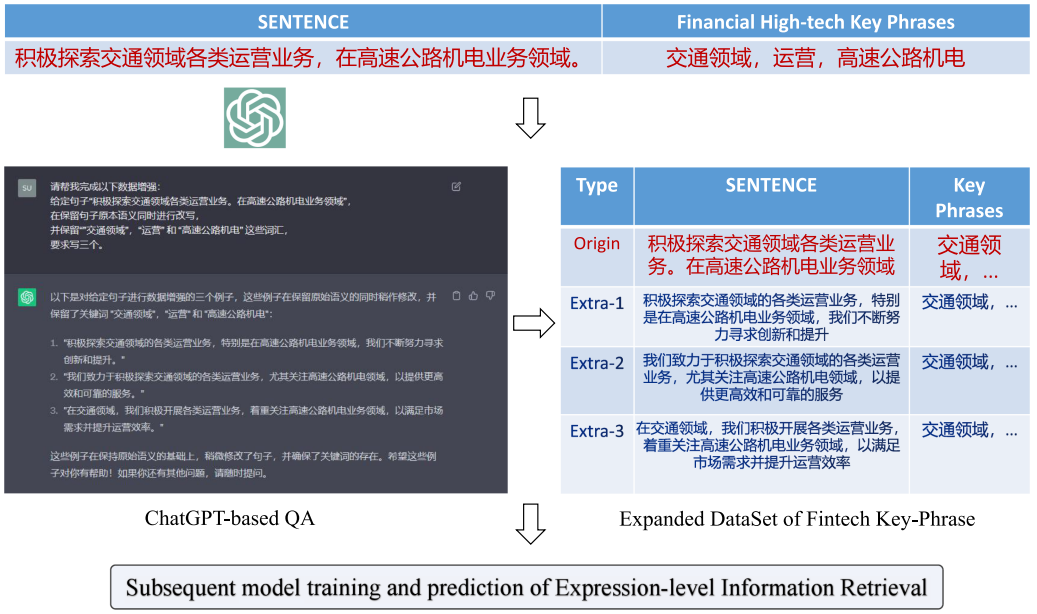


Fig. 13. An example of text augmentation using ChatGPT based on the Fintech Key-Phrase dataset. We use the ChatGPT API interface for efficient and batch processing, rather than utilizing a UI interface, which was just for conceptual visualization.

this text augmentation strategy significantly improve the performance of expression-level information extraction tasks in the high-tech financial domain, in terms of the coverage and accuracy of extracted phrases.

It is widely acknowledge to us that prompting engineering is a cumbersome art that requires abundant experience and manual trial-and-errors [19, 45]. To design the suitable prompts to trigger the rich prior knowledge and conversational capability of ChatGPT, we manually elaborate several prompt templates following the general communication habits. Furthermore, we insert two types of slots into each designed prompt template to combine the corresponding mentions, which, respectively, represent `[SENT]` (slot of sentence) and `[Key-Phrase-i]` (slot of $i-th$ annotated phrase) for experimental simplicity. In order to conduct the subsequent experiments, we collaborated with financial and linguistic experts to evaluate the quality of each template using multiple question-answer sessions with ChatGPT. Their expertise proved valuable in selecting the best-performing prompt template from a pool of candidates. The chosen prompt template is presented below.

> 请帮我完成以下数据增强: \n 给定句子 **[SENT]**, \n 在保留句子原本语义同时进行改写, \n 并保留 **[Key-Phrase-1]**, **[Key-Phrase-2]**, ..., 和 **[Key-Phrase-N]** 这些词汇, \n 要求写三个.

Subsequently, we utilized a script to invoke the official ChatGPT API[23] and performed data augmentation on the Fintech Key-Phrase training set. Through interactive sessions with ChatGPT, we obtained a batch of responses and extracted these expanded text to the original Fintech Key-Phrase dataset. Based on this, we obtained an additional 34,752 augmented texts from the

[23]ChatGPT API address: https://api.openai.com/v1/engines/davinci-codex/completions

training set, which is approximately equivalent to 3.2 times the size of the original training set of Fintech Key-Phrase. Subsequently, the augmented texts were integrated with the original dataset to create an enhanced version of Fintech Key-Phrase that had undergone text augmentation. The corresponding expanded dataset has been made publicly available on Github.[24]

Furthermore, to ensure the quality and reliability of our ChatGPT-based data augmentation, we have taken proactive measures towards the augmented sentences. Recognizing the potential issue of unreliable sentences, such as hallucinations, associated with large language models, we implemented a rigorous quality control process. This involved human reviewers meticulously examining and scrutinizing each generated sentence to detect and rectify any anomalies, ensuring that our augmented dataset maintains a high level of credibility. While this manual review process was undoubtedly time and resource-intensive, we believe that the effort invested in mitigating the possibility of language model-generated hallucinations and other potential noise-related issues is entirely justified. The meticulous review process safeguards the integrity of the Fintech Key-Phrase and reinforces the robustness of our data augmentation outcomes.

Based on the same experimental setup, we conducted performance evaluation experiments on all baseline models using the augmented Fintech Key-Phrase dataset. Note that only the training set underwent text augmentation, while all other settings remained unchanged, including various hyperparameters and the test set. Figure 14 illustrates the performance changes of the models in the information retrieval task before and after text augmentation. From the figure, it can be observed that after applying ChatGPT-based data augmentation strategy, all SOTA models experienced a significant improvement in keyword extraction accuracy. Furthermore, in multi-turn training, the baseline models also showed some improvement in the maximum and minimum metrics related to random perturbations, with an overall accuracy increase of approximately 1% to 3%. For example, the BERT-Linear and RoBERTa-BiLSTM-CRF baselines demonstrated notable performance improvements, with an absolute accuracy increase of approximately 3%. The comprehensive comparative experiments clearly demonstrate the effectiveness and superiority of this text augmentation strategy in keyword extraction for the finance and high-tech domains.

To cater to the distinct requirements of various users, we have incorporated a additional feature into our dataset—a flag function that clearly distinguishes between human-annotated instances and ChatGPT-based instances. Each data point in our dataset is now marked with a flag indicating its origin, whether it is a human-authored text or a generated virtual text. This enhancement ensures that users can readily identify and utilize the data that aligns with their specific needs and preferences. We believe that this clear demarcation not only enhances the usability of the dataset but also promotes transparency in the data-sharing process, fostering a more informed and tailored approach to research and application.

In conclusion, the utilization of ChatGPT for text augmentation offers several benefits for improving the expression-level information extraction performance. It significantly enhances key phrase extraction accuracy in SOTA models and improves their robustness in handling random perturbations during multi-turn training. Additionally, the text augmentation process maintains consistency in model settings, ensuring a fair comparative evaluation. This strategy's effectiveness extends beyond finance and high-tech domains, making it versatile for enhancing model performance across other different NLU tasks. Overall, ChatGPT-based text augmentation improves the performance on Fintech Key-Phrase, increases model robustness, and demonstrates efficacy in key phrase extraction tasks across various domains.

[24]Augmented Fintech Key-Phrase: https://github.com/albert-jin/Fintech-Key-Phrase/blob/main/Fintech-Key-Phrase/data_augment

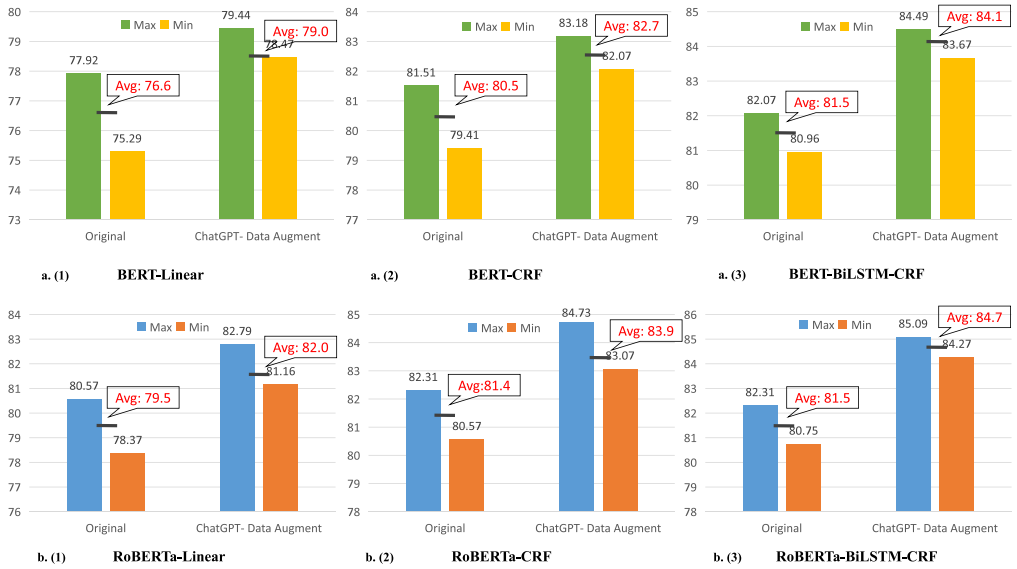


Fig. 14. The performance comparisons (Accuracy %) between the dataset before and after data augmentation that were evaluated on the original test set of Fintech Key-Phrase. Rows *a.* and *b.* indicate the PLMs employments of BERT and RoBERTa, respectively.

6 RELEASED TOOLS FOR FINANCIAL HIGH-TECH DOMAIN INFORMATION RETRIEVAL

To motivate newcomers interested in the financial high-tech specific-domain information retrieval field, we have developed a series of functional tools. These tools include a website,[25] which provides brief introductions on our Fintech Key-Phrase and features six real-time user interactive planes for information extraction. The website can be accessed at https://albert-jin.github.io/FintechKP-frontend/model_prediction_rblc.html. The information extraction is based on six adopted baselines, and the corresponding APIs for information retrieval can be accessed at https://31863ew564.zicp.fun/information_retrieval. Our tools enable users to learn more about our *Fintech Key-Phrase* benchmark and query key expressions from Chinese financial high-tech domain text. Below are the details of the released tools.

6.1 Website Of Our Fintech Key-Phrase

Figure 15 illustrates the homepage of the website https://albert-jin.github.io/FintechKP-frontend/, which consists of four major sections:

- **Introduction:** This section gives a brief overview of the *Fintech Key-Phrase* benchmark. (1) The main plane displays the original statistics and the train/test split details of the *Fintech Key-Phrase* dataset. (2) By clicking on the '*Learn More*' button, users can access specific information about the *Dataset Construction Motivation*, and *Dataset Construction Procedures*;
- **Model Prediction:** This section offers convenient user experiences by showcasing the performance of well-trained baselines for all users. In the *Real-time Model Prediction* section, users can click on one of the six sub-planes (e.g., the *BERT-BiLSTM-CRF* or

[25] Online Website: https://albert-jin.github.io/FintechKP-frontend/

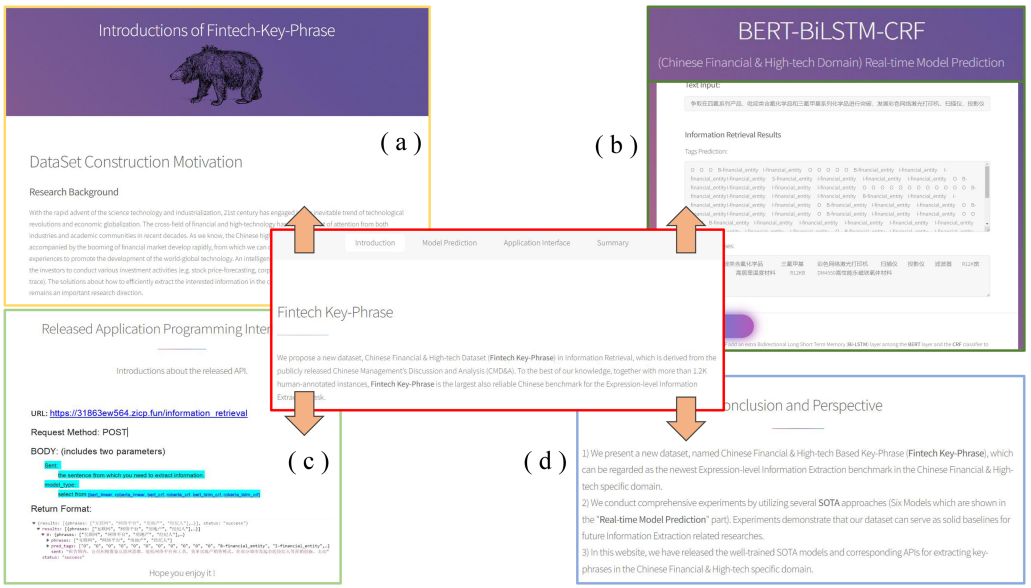


Fig. 15. The functional partitions of our released website, with (a), (b), (c), and (d) representing the four sub-pages: Introduction, Model Prediction, Application Interface, Summary, respectively.

RoBERTa-BiLSTM-CRF) to access a new user interactive screen. For example, clicking on the sub-plane *RoBERTa-BiLSTM-CRF* take users to the real-time information retrieval functions powered by the *RoBERTa-BiLSTM-CRF* baseline. Users can then enter a piece of Fintech-domain text into the sentence input component and click the '*Search*' button to request the loaded *RoBERTa-BiLSTM-CRF* model to predict all possible key-phrases from the backend system. The system returns the queried results, and the corresponding tagging results and extracted key-phrases are displayed in the textareas 'Tags Prediction' and 'Predicted Phrases', respectively;

— **Application Interface:** This section provides detailed information about API requesting and calling approaches for each well-trained baseline. It includes various calling parameters such as headers, body, and standard returned formats;

— **Summary:** Finally, in the last part of the website, we summarize the major contributions of our proposed 'Fintech Key-Phrase' benchmark.

6.2 Released APIs

To facilitate future research on domain-specific information retrieval, we have released a series of online APIs. The APIs can be accessed at https://31863ew564.zicp.fun/information_retrieval for interested users. As shown in Figure 15(c), users need to prepare the necessary request parameters in advance, which include:

(1) The request method should be set to 'POST' as it is the only allowed method. Please note that the API querying process may introduce a certain network delay, so it is advised to wait patiently for a few seconds;
(2) The request headers should include 'Content-Type: application/json' to ensure that the returned data adheres to the standard JSON format;
(3) The body parameter 'sent' should contain the complete Chinese sentence. Additionally, please select a baseline from the provided list: [bert_linear, roberta_linear, bert_crf,

Table 6. Case Analysis of the Information Extraction Performance Comparisons between Our 'RoBERTa-BiLSTM-CRF' Baseline and the *TENCENT TextSmart* Official Released API

Original Text	Expert Annotations	RoBERTa-BiLSTM-CRF Trained on FintechKP	TexSmart Preds
从长期来看，全球天然气的需求将增长50%以上，是化石燃料中增长最快的。	天然气 化石燃料	天然气 化石燃料	长期 50% 化石燃料
研究院下设聚氨酯研究所、醇胺工程研究所、节能新材料研究所、分析检测中心等部门。	聚氨酯研究 醇胺工程研究 节能新材料研究 分析检测中心	聚氨酯研究 醇胺工程研究 节能新材料研究 分析检测中心	聚氨酯研究 醇胺工程研究所 节能新材料研究所 分析检测中心
单抗行业现状及未来发展趋势单克隆抗体药物是当今国际医药界的前沿领域。	单抗 单克隆抗体药物	单抗 单克隆抗体药物	未来 当今
植物提取物、人参提取物、人参皂苷单体分离方法的研究、产品的开发取得了阶段性成果。	植物提取物 人参提取物 人参皂苷单体分离	植物提取物 人参提取物 人参皂苷单体分离	植物提取物 人参提取物 人参皂苷单体分离
其中3个品种获得批准，罗库溴铵注射液为全国首家通过一致性评价。	罗库溴铵注射液	罗库溴铵注射液	罗库溴铵注射液 3个
云南省盐资源储量最大、生产成本最优的四个盐矿（昆明盐矿、一平浪盐矿、乔后盐矿和普洱制盐分公司）已是本公司的生产企业。	盐资源, 盐矿 昆明盐矿 一平浪盐矿 乔后盐矿 普洱制盐	盐资源, 盐矿 昆明盐矿 一 平浪盐矿 乔后盐矿 普洱制盐	盐资源, 盐矿 云南省, 盐, 四个 平浪盐矿, 乔后盐矿 昆明盐矿, 一平浪盐矿 普洱制盐分公司

Referring to the expert annotations as standards, words marked in red and underlined indicate inaccurate predictions, and words marked in green denote missing predictions.

roberta_crf, bert_lstm_crf, roberta_lstm_crf], and attach it to the body parameter 'model_type';

(4) Furthermore, the real-time APIs are continuously maintained for long-term usage. We encourage interested users to proactively contact us and report any questions or issues by sending an e-mail to weiqiangjin@stu.xjtu.edu.cn.

7 CASE STUDY

To gain a deeper understanding of the significance of our proposed *Fintech Key-Phrase*, we collect a comparison of information retrieval instances between our well-trained baseline 'RoBERTa-BiLSTM-CRF' and the *TENCENT TextSmart* tool.[26] *TENCENT TextSmart* is a well-known and convenient open-domain information retrieval tool in China. Table 6 presents four specific examples, with each case displaying the standard results of expert annotations, our trained baseline's extracted phrases, and the "*TexSmart*'s extracted phrases. As we expected, all the samples clearly demonstrate the effectiveness of our 'RoBERTa-BiLSTM-CRF', which is fully trained on *Fintech Key-Phrase* benchmark. It is worth noting that the experts focus primarily on concept-level phrases in the financial high-tech domain rather than tangible objects.

Based on our observations, the following conclusions can be drawn:

(1) In the first case (second row), our trained baseline accurately predicts all the expert-annotated phrases, while the *TexSmart* API makes several incorrect predictions, including redundant extracted expressions ('长期', '50%') and missed expressions ('天然气').

[26]TexSmart semantic understanding tool: https://texsmart.qq.com/en

(2) In the second case (third column), our model performs well, but the *TexSmart* exhibits minor errors. Compared with the expert annotation phrases, these errors include missing the words '研究' for the phrase '聚氨酯研究'; over-predicting the word '所' for the phrases, '醇胺工程研究' and '节能新材料研究'; and missing the words '分析' for the phrase '分析检测中心'.
(3) In the third case (fourth column), *TexSmart* provides completely wrong predictions: missing two key-phrases ('单抗, 单克隆抗体药物') and making redundant predictions ('未来, 当今'); The comparisons with *TexSmart* clearly demonstrate the superior accuracy of our model's predicted results.
(4) In the fourth case (fifth column), our RoBERTa-BiLSTM-CRF model trained on Fintech Key-Phrase is able to predict all three phrases, but *TexSmart* fails to capture the key words '单体分离' in the key phrases '人参皂苷单体分离';
(5) In the fifth case (sixth column), our fully-trained RoBERTa-BiLSTM-CRF model is capable of predicting the phrase '罗库溴铵注射液' (Rocuronium Bromide Injection), whereas TexSmart fails to provide the correct answer. Instead, it excessively predicts a meaningless term, '3个'.
(6) In the sixth case (last column), our model fully trained on the Fintech Key-Phrase dataset achieved accurate extraction results, with only minor omissions. It missed the term '昆明' in the phrase '昆明盐矿' (Kunming Salt Mine) and the term '一' in the phrase '一平浪盐矿' (Yipinglang Salt Mine). Overall, the model's results align closely with expert annotations. In contrast, TexSmart overlooked most of the core phrases, such as '盐资源' (salt resources), '平浪盐矿' (Pinglang Salt Mine), '乔后盐矿' (Qiaohou Salt Mine), and '一平浪盐矿' (Yipinglang Salt Mine). This starkly contrasts with our expectations.

The presented cases provide substantial evidence that our fully-trained 'RoBERTa-BiLSTM-CRF' outperforms other methods in expression-level information retrieval tasks in the financial high-tech domain. In conclusion, the case analysis demonstrates the practicality and usefulness of providing a high-quality dataset to address the accuracy issues in domain-specific information extraction tasks.

8 CONCLUSION

In this work, we propose the Fintech Key-Phrase dataset, a high-quality human-annotated benchmark that is the first and largest Chinese corpus specifically designed for information extraction in the financial high-tech field. The dataset fills the gap in training data sources in the Chinese financial high technology domain's information retrieval task. We have followed meticulous annotation procedures based on predefined guidelines to ensure the overall quality of the dataset. Furthermore, we conducted comprehensive noise evaluations for Fintech Key-Phrase, which included assessing annotator consistency and the accuracy of annotations. This evaluation provided valuable insights into the dataset, enhancing our understanding of its characteristics. We hope that our proposed corpus extraction methodology and key-phrase annotation guidelines will be valuable for future research in domain-specific information retrieval. To establish solid baselines for future studies, we have conducted comprehensive experiments using several SOTA approaches. It is of significance to mention that inspired by the prominent text generalization ability of ChatGPT, we firstly introduced ChatGPT in the data augmentation process of Fintech Key-Phrase. Using ChatGPT, we expanded the original dataset by over three times. Through extensive comparative experiments, we verified the effectiveness of this text augmentation strategy in improving the information retrieval performance. The Fintech Key-Phrase benchmark and corresponding experimental codes are available at https://github.com/albert-jin/Fintech-Key-Phrase. Additionally, we have developed a customized website https://albert-jin.github.io/FintechKP-frontend/ and

real-time information retrieval APIs https://31863ew564.zicp.fun/information_retrieval for the Fintech Key-Phrase benchmark, aiming to facilitate future research in the Chinese financial high-tech information retrieval task.

In the future, we have planned the following works: (1) Annotate more financial high-tech domain corpora from other potential information channels (such as Chinese Business Website[27] and Innovation China[28]) to further explore the potential of neural networks; (2) Conduct additional experiments using advanced information retrieval approaches on our Fintech Key-Phrase dataset to further enhance performance; (3) Explore the utilization of well-trained models on the Fintech Key-Phrase dataset to facilitate and optimize other downstream NLP tasks.

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AUTHOR CONTRIBUTION

Weiqiang Jin: Conceptualization, Software, Investigation, Methodology, Formal analysis, Visualization, Project administration, Writing - original draft, Writing - review & editing. **Biao Zhao:** Investigation, Writing - review & editing, Validation. **Yu Zhang:** Methodology, Visualization. **Gege Sun:** Supervision, Formal analysis. **Hang Yu:** Writing - review & editing, Funding acquisition.

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